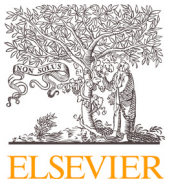
Journal of Economic Dynamics & Control 183 (2026) 105247



Contents lists available at ScienceDirect

Journal of Economic Dynamics and Control

journal homepage: www.elsevier.com/locate/jedc





Bank lending standards and the U.S. economy ⋆

Elijah Broadbent [a], Huberto M. Ennis [b], Tyler J. Pike [c], Horacio Sapriza [b,*]

[a] Ohio State University, USA
[b] Richmond Fed, USA
[c] University of Maryland, USA

ARTICLE INFO

JEL classification:
E32
E44
G21

Keywords:
Credit supply
Macroeconomic activity
Loan portfolio composition

ABSTRACT

The provision of bank credit to firms and households affects macroeconomic performance. We use survey measures of changes in bank lending standards, disaggregated by loan category, to quantify the effect of changes in banks' attitudes toward lending on aggregate output, inflation, and interest rates. Bank lending to businesses is particularly important for macroeconomic outcomes, with peak effects on output growth of around 30 to 40 basis points after four quarters of the initial shock. These effects depend on the stage of the business cycle and the proximity of the short-term interest rate to its effective lower bound. The effects are larger when output is growing below trend, when uncertainty is increasing or high, and when interest rates are away from the lower bound. We also find that the response of the economy to lending-standards shocks is asymmetric, with tightening shocks having larger effects on output.

1. Introduction

Bank lending conditions are generally considered an important factor when evaluating the prospects for macroeconomic activity and the stance of monetary policy. During the July 26, 2023, post-FOMC press conference, Fed Chair Powell stated, "*The SLOOS has been telling us for more than a year that banking conditions are tightening. That process is ongoing, and that will restrain economic growth.*" A few minutes later, he underscored the multidirectional (endogenous) nature of the effects when he pointed out that "*an expected result of tightening interest rate policy is that bank lending conditions would tighten as well.*" In this paper, we aim to better understand the role that changes in bank lending standards play in the U.S. economy, explicitly accounting for possible feedback effects, i.e., endogeneity.

There is broad agreement in economics about the critical importance of financial markets and institutions for macroeconomic outcomes. Banks, in particular, play a highly consequential role in the provision of credit to the U.S. economy. Accordingly, the literature has dedicated considerable attention to how banking conditions affect aggregate economic activity (e.g., see Bernanke, 1983, for a classic reference and Brunnermeier et al., 2013, for a good survey).

⋆ We thank participants in seminars at the Federal Reserve Board and the IMF, and the audiences at the 2019 Medellin IFABS, the 2024 MidWest Macro, LAMES/LACEA, and Econometric Society meetings. We are also grateful to Andrea Ajello, Paul Ho, Felicia Ionescu, Christian Matthes, Camelia Minoiu, and Francisco Vazquez-Grande for their comments. Larissa Chan, Kushal Patel, and Alex Tan provided excellent research assistance. The views expressed here are solely those of the authors and do not necessarily represent the views of the Federal Reserve Bank of Richmond or the Federal Reserve System.

* Corresponding author.
E-mail addresses: broadbent.24@buckeyemail.osu.edu (E. Broadbent), huberto.ennis@rich.frb.org (H.M. Ennis), tjpike@umd.edu (T.J. Pike), horacio.sapriza@rich.frb.org (H. Sapriza).

https://doi.org/10.1016/j.jedc.2025.105247
Received 10 March 2025; Received in revised form 18 December 2025; Accepted 19 December 2025
Available online 30 December 2025
0165-1889/© 2025 Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

The banking sector remains a significant portion of the U.S. financial system. At the end of 2019, 15% of assets in the domestic financial sector were located on the balance sheets of U.S. chartered depository institutions, with the participation increasing to 40% when considering the provision of loans to businesses and households. While these proportions tend to be lower relative to other advanced economies, they are still very significant; for many economic actors, banks are their primary and, often, exclusive source of funding.

How important for the aggregate economy are changes in the provision of bank credit? This is a challenging question to answer because lending depends on multiple forces. Furthermore, feedback effects are likely to play an important role, as suggested by Fed Chair Powell's quote above: not only do banks influence the economy, but also the state of economic activity, and of monetary policy, impact the way banks evaluate lending opportunities (Brunnermeier et al., 2021). Therefore, identifying how banks' attitudes toward lending affect the performance of the economy requires careful empirical analysis.

In this paper, following Lown and Morgan (2006) and Bassett et al. (2014), we use canonical time-series techniques to empirically study how changes in banks' lending standards affect economic activity, inflation, and interest rates. We go beyond these seminal papers by providing a comprehensive investigation of the origins of the macro effects from changes in bank lending standards. We make three distinct contributions: first, taking a more disaggregated data approach, we determine the relative importance of lending standards for businesses and households, separately, in driving the identified macro effects. Second, we study whether the responses depend on the state of the economy: in particular, the stage of the business cycle, the proximity of interest rates to their effective lower bound, the evolution of economic uncertainty, and the direction of change in banks' attitudes toward lending. Third, we provide a thorough reassessment of the relative advantages of using micro data to identify macro effects of changes in lending standards. We think this is an important step because access to the micro data is restricted but, as we show, is often not strictly required for advancing the study and understanding of these topics.

For several decades now, the Federal Reserve has been consistently collecting survey data about banks' attitudes toward lending. The Senior Loan Officers Opinion Survey on Bank Lending Practices (SLOOS) is a quarterly survey that collects information on lending standards, loan terms, and loan demand, to businesses and households across loan categories. The survey defines *lending standards* as the processes that banks follow for approving or denying loan applications. *Loan terms* are the specific conditions that apply to those loan contracts that have already been approved. Aside from credit spreads and fees, the terms of loans include covenants, collateral requirements, and credit and maturity limits. In that way, changes in lending standards tend to capture variations in the extensive margin of lending, while changes in loan terms are more closely related to the intensive margin. Information on *loan demand* in the SLOOS captures fluctuations, as perceived by surveyed banks, in their customers' demand for loans.

In principle, banks could adjust the pricing of loans (i.e., terms) to manage their lending activity. Within the context of imperfect information, however, it has been long understood that interest rates are not sufficient to optimize the general lending decision (Stiglitz et al., 1981; Williamson, 1987). Indeed, banks use multiple levers in the process of adjusting their lending—with, for example, collateralization requirements such as loan-to-value ratios playing an important role in many contexts. We focus on the changes in lending standards required by banks to approve loan applications, a key tool in loan-related decision making. In our empirical analysis, we include direct measures of interest rates and risk premia, as commonly reported in economic data, instead of using more qualitative survey information about these variables, such as the questions about loan terms included in the SLOOS.[1]

More broadly, we are interested in understanding how changes in the supply of bank credit affect the macroeconomy. Because the supply of credit is likely to respond to many of the same factors that drive the demand for credit (and economic activity more generally), there is a risk of simultaneity bias when trying to identify the effects of changes in credit supply. Furthermore, as mentioned before, causation may run both ways, from lending standards to economic conditions and from economic conditions to lending standards. To address these issues, we use multiple identification methods, which allows us to contrast and compare results and discern which are the most robust.

Conveniently, the SLOOS includes bank-level loan demand information that can be helpful in the process of sorting out demand and supply effects. Following the literature, we use the loan demand data in the survey to help us identify exogenous changes in lending standards (net of effects that can be attributed to common factors affecting both demand and supply of bank loans, including reactions to changes in economic conditions). These exogenous changes allow us to describe, using standard tools such as impulse response functions, how key macroeconomic variables in the U.S. are impacted by changes in the supply of bank credit. While, ultimately, we are interested in assessing *all* direct and indirect macroeconomic effects of changes in the provision of bank credit, to disentangle those effects from other confounding factors, it is critical to be able to identify exogenous changes to lending standards.

The first step in our empirical analysis is to construct diffusion indexes for bank lending standards and bank loan demand for the sample period that goes from 1996 to 2023. We also use cross-sectional data from the SLOOS, combined with bank-level balance sheet data from the Call reports, and several macroeconomic variables, to construct a time series of aggregate *exogenous* shocks to bank lending standards for the main categories of bank loans. In doing so, we follow the pioneering work of Bassett et al. (2014) (see also Cavallo et al., 2024b).

We then perform two main empirical exercises to assess the impact that changes in bank lending standards have on macroeconomic performance—as captured by output growth, interest rates, and inflation. First, we specify a small-scale vector autoregressive (VAR) model and analyze the corresponding impulse response functions and the forecast error variance decomposition. We consider both a measure of overall lending standards and a specification that includes standards for loans to households and to businesses, separately.

[1] Relatedly, Lown and Morgan (2006) find that direct measures of the average interest rate on bank loans do not significantly increase explanatory power once lending standards have been taken into account.

We find that a tightening of overall lending standards reduces real GDP growth in the two years after the shock, with a peak effect after one year. We also see that bank lending growth drops quickly in the quarters after the shock, which suggests a transmission channel from lending standards, to bank lending, to GDP growth. To dig deeper into the mechanisms at play, we look at shocks to lending standards for businesses and for households separately and show that only the former has significant effects on GDP growth and bank lending. In that sense, the effects we associate with overall lending standards seem to reflect mainly changes in lending standards *to businesses*.

Our variance decomposition analysis confirms that business lending standards are three times more important than household lending standards for explaining the variance in real GDP growth. Business lending standards also account for an important portion of the variance of the federal funds rate, particularly at long time horizons. Additionally, we find significant feedback effects, where real GDP growth explains a significant amount of the variation in business lending standards, and less so of household lending standards.

To further discern what drives the effects of changes in lending standards on the macroeconomy, we conduct a second exercise which accommodates the possibility of asymmetries in the responses, conditional on the state of the economy. Specifically, we use *Threshold VARs* (TVARs) to accommodate the possibility that the effects depend on various partitions of the state of the economy: (1) the state of the business cycle, as measured by the level of GDP relative to trend; (2) the level of the interest rate and its proximity to its effective zero lower bound; (3) the evolution of economic uncertainty; and (4) whether the economy experiences a tightening or an easing shock to lending standards. We find that the effects of a tightening shock to standards are significant when GDP is below trend, when interest rates are away from their effective lower bound, and when uncertainty is increasing or high. It is also the case that only tightening lending standards have a clear-cut and significant effect on aggregate output, while the effect on inflation is more nuanced.

Overall, we conclude that lending standards meaningfully affect bank credit and GDP growth, and these effects are mainly driven by the response of the economy to *tightening* in lending standards *to businesses* when the level of GDP is below trend, short-term interest rates are away from the effective lower bound, and uncertainty is increasing or high. While the effect on inflation is relatively muted, we do observe a tendency of interest rates to move to accommodate (and counteract) the tightening in standards (by falling), but this is naturally most operative when the interest rate has room to fall by being away from its lower bound.

The rest of the paper is organized as follows. The next paragraphs provide a brief review of the literature, and Section 2 describes the data. Section 3 explains the construction of the diffusion indexes, how we use the cross-sectional data to produce a sequence of exogenous lending-standards shocks, and a description of the VAR specifications we use in the empirical exercises of the following sections. Section 4 studies how shocks to lending standards in the business and household sectors affect the macroeconomy, and Section 5 studies potential state-dependent and asymmetric effects. Section 6 investigates several important robustness checks, and Section 7 concludes.

Related Literature: Our analysis builds on the work by Lown and Morgan (2006) and Bassett et al. (2014). The construction of indexes for bank lending standards incorporates enhancements introduced by Glancy et al. (2020) and used regularly at the Federal Reserve Board to describe the evolution of bank lending conditions.

Using SLOOS data, Castro et al. (2022a) study idiosyncratic changes in credit supply at the individual-bank level and how these changes affect the provision of loans by the bank in the quarters and years after that change.[2] They study C&I lending and find that a one-standard-deviation increase in their index (a tightening move) is associated with a 3.13 percentage point drop in C&I loan growth in the following year. An important distinction with our analysis is that they consider the impact of a move to tighter credit by a given bank —relative to other banks in the same period (quarter). Here, instead, we focus on the effects on the macroeconomy of a *general* (across all banks) move to tighter lending standards

Cavallo et al. (2024b) take the micro approach to extracting exogenous shocks to bank lending standards and apply it to the different loan categories, as we do here. We go a step further and group the subcategories into broader categories based on statistical properties of the series. This allows for a more parsimonious approach which we can then use to assess the relative importance of household and business lending standards in driving the aggregate effects. In related work, Cavallo et al. (2024a) use local projections to explore the effects of different categories of lending standards on the components of aggregate demand (PDFP), while we consider the joint effect on a set of macroeconomic variables, including output, inflation, and interest rates. Moreover, and importantly, we also look at nonlinearities and asymmetries of the response of the macroeconomy, which they do not investigate.

There is a closely related empirical and theoretical literature aimed at understanding the endogeneity of credit standards. Two recent examples are Jimenez et al. (2012) and Swarbrick (2023). Using data from the Spanish credit registry, Jimenez et al. (2012) study empirically how financial and economic conditions affect the willingness of banks to accept loan applications. They find that bank balance-sheet strength, in the form of liquidity and capital, play a meaningful role in that process.[3] Swarbrick (2023) sets up a dynamic macroeconomic model where informational frictions introduce a role for probabilistic approval of loan applications. In the model, loan risk drives changes in lending standards and tends to amplify the effects of credit-risk shocks on economic activity. This is in line with evidence uncovered by Castro et al. (2022b) using SLOOS data for C&I lending, combined with indicators of loan performance. We share with these papers the ultimate objective of understanding the importance of bank credit for economic activity.

[2] Instead of using only lending standards, these authors construct a summary index that reflects changes in both lending standards and loan terms collected in the SLOOS.

[3] Chodorow-Reich (2014) also finds that firms' relationships with banks and the financial conditions of banks matter for understanding the flow of bank credit to firms in the U.S., especially smaller firms. See also Baslandze et al. (2023) for a recent estimate of the impact of credit supply shocks on net job creation by small and medium-sized firms.

Here, however, the main focus is on *exogenous shocks* to credit standards as a way to address possible simultaneity bias and, in that way, disentangle the net impact that changes in bank credit supply can have on the economy.

There is also a large and growing empirical literature that uses bank, firm, and sometimes loan-level data to study the importance for economic activity of bank funding (e.g., Chodorow-Reich et al., 2022; Ivashina et al., 2022; Alfaro et al., 2021; Jiménez et al., 2020). Relative to our paper, this literature takes a purely microeconomic perspective on the possible mechanisms behind some of the macroeconomic effects identified in our study.

While our focus here will be on bank lending, there is an extensive literature studying the impact that the overall supply of credit in the economy can have on macroeconomic outcomes. A good recent example of these efforts is Gambetti and Musso (2017), who estimate a structural time-varying VAR model with stochastic volatility using data for the U.S., U.K., and Europe. Instead of lending standards, as we use here, their data include loan volumes (not just from banks) and a measure of the average lending rate. This allows them to use a quarterly time series going back to 1980. Their identification strategy is also different from ours, as they use sign restrictions to identify various shocks according to predictions from theory. Similar to what we find here, Gambetti and Musso (2017) find that their identified credit supply shock impacts bank lending and GDP growth, although the responses they find for the U.S. are shorter-lived than the ones we estimate.

Mumtaz et al. (2018) refines the use of sign restrictions for identification and for understanding the impact of credit supply shocks on the macroeconomy. They also caution the interpretation of lending standard shocks as proxies for general credit supply shocks. In their study, using sign restrictions and forecast error variance decomposition to identify a general credit supply shock in the structural VAR, they find that a shock that increases credit spreads by 10 basis points has a large and persistent negative impact on the economy, with credit growth declining by 0.8% and investment, consumption, and output growth all declining by meaningful amounts (2%, 0.4%, and 0.5%, respectively).

An important contribution to this broader empirical literature is the paper by Barnichon et al. (2022), which focuses on asymmetries in the response of the economy to positive and negative financial shocks estimated using corporate bond prices. Specifically, these authors allow the empirical model to respond differently to favorable and unfavorable financial shocks, and find that the latter have stronger and more persistent effects on the economy than the former. Our results are consistent with their findings in that our threshold VARs, by breaking the linearity in the standard specification, allow us to verify that tightening shocks to lending standards are a lot more consequential for GDP growth than easing shocks.

Forni et al. (2024) also investigates empirically the impact of general financial shocks on the macroeconomy and the possible nonlinearities in those responses. Their main finding is that financial shocks have to be negative and large to meaningfully affect the aggregate economy. While we are able to verify that negative shocks are more important than positive shocks in our context, data availability limits our ability to separate shocks into both size and direction categories. We do revisit their finding that economic uncertainty matters for the effect of financial shocks in the economy and find results that go in line with theirs.[4]

2. The data

Our measure of lending standards comes from the SLOOS. The Federal Reserve conducts this quarterly survey to monitor lending conditions in the banking sector. The survey has been in place, in different forms, since 1964. However, for the purpose of this paper, we restrict attention to the period between 1996 and 2023, for which the data are available at the level of disaggregation that we are interested in exploiting here.

The SLOOS panel consists of a sample of up to 80 U.S. domestic commercial banks selected to be geographically diverse, mutually independent, and representative of a high proportion of the lending volume in each of the major bank lending areas.[5] Towards the end of our sample period, close to 50 percent of assets in domestically chartered commercial banks were accounted for by SLOOS respondents, and each participating bank held at least two billion dollars in total assets. While bank participation is voluntary, response bias is unlikely to be a major limitation of the SLOOS as banks invited to join the panel nearly always agree to participate and, once included, consistently complete the survey.

We also compute the share of loans in the SLOOS banks that is accounted for by the top 15 banks by assets. This share increased steadily in the first half of our sample period, from 50 percent in 1996 to close to 80 percent in 2012, but has been gradually trending down since then, to around 65 percent at the end of our sample period. This pattern partly reflects the general consolidation trends in the U.S. banking system, with the largest banks gaining share during the 2000s and later leveling off in the 2010s (McCord et al., 2014).[6]

The SLOOS survey asks participating banks to report changes to lending standards during the survey period, by category of loans, using the following general template:

"Over the past three months, have your bank's credit standards for loans in category k *changed?"*

[4] Another set of papers in this general body of work dedicated to studying credit supply shocks is the one that exploits structural theoretical models to quantify the role of financial frictions and shocks on U.S. macroeconomic performance. A good example is Ajello (2016).

[5] The SLOOS survey also contains information from U.S. branches and agencies of foreign banks. Following the previous literature, we do not include those responses in the main portion of our investigation.

[6] See the online supplementary material for more information about the data and other issues throughout the paper.

Banks answer these questions subjectively using a scale from 1 to 5, with lower numbers indicating movement in the direction of tightening lending standards, and higher numbers corresponding to easing. The middle of the scale indicates that standards remain mostly unchanged over the survey period. Because banks rarely characterize changes in standards using the most extreme labels on either end, we redefine directional variables for each loan category k without distinction of intensity,

$$I_{it}[k] = \begin{cases} -1 & \text{if bank } i \text{ reported easing standards for category } k \text{ in quarter } t, \\ 0 & \text{if bank } i \text{ reported no change in standards for category } k \text{ in quarter } t, \\ 1 & \text{if bank } i \text{ reported tightening standards for category } k \text{ in quarter } t. \end{cases}$$

Note that, relative to the survey, this definition not only reduces the grading from five to three, but also effectively inverts the scaling so that increases in the lending standards index indicate tightening. This inversion is mainly done for expositional reasons, following the literature.

Each loan category in the data belongs to one of four main *groups of categories*: C&I, CRE, Consumer Credit, and RRE. We construct bank-level weights for each category using loan data from the call reports, which are collected quarterly for supervisory purposes. We also use call-report data to construct an *adjusted* series of lending standards, as we explain below.

For the estimation of our medium-size macroeconomic VARs, we use standard variables such as real GDP growth, unemployment, inflation, and the real federal funds rate. In some specifications, we also include a measure of the credit spread developed by Gilchrist and Zakrajšek (2012) (GZ) and a measure of growth of core lending capacity for the banking system.[7]

A quick look at the data makes clear that the COVID period is an outlier in macroeconomic terms. For this reason, in our benchmark estimation, we use data up to 2019:Q4. We could instead deal with the COVID period by implementing the volatility correction proposed by Lenza and Primiceri (2020). However, as these authors explain, if inference is the sole goal of the VAR estimation—as it is in our study—then simply excluding the COVID shock from the sample will have the same effect as their alternative approach, without introducing extra noise. In the robustness section, we present the estimation including the COVID period.

Before we describe the empirical design, it is worth highlighting a few differences with the data used in Bassett et al. (2014), which we intend to evaluate and extend. First, our sample period starts in 1996:Q1, instead of 1991:Q4, and goes all the way to 2023:Q4, instead of only until 2012:Q3. The specific questions about standards for consumer lending are available beginning in 1996:Q1, which is why our sample period starts there. Furthermore, the level of disaggregation by categories within the main groups of loans in the SLOOS data only becomes significant later in the period. Given that we add many more years at the end of the sample period, and to the extent that the methodology pursued here is intended to exploit such disaggregation, it makes sense to start the sample period for this study in 1996.

Another relevant difference with the data employed in Bassett et al. (2014) is that we use responses by all domestic banks in the survey instead of restricting to banks that are publicly traded. As a result, we are not able to use stock valuations in our study (which they do), and we instead use profitability measures that are available for all banks in the sample. As part of the robustness exercises in Section 6, we restrict our sample to the group of largest banks (all of them publicly traded), and show that results are similar.

3. Lending standards and their effects

The first step in our empirical strategy is to construct an aggregate index of changes in lending standards in the banking system. For this, we use the responses of individual banks appropriately weighted by the proportion of total bank lending that they hold in their respective balance sheets.

We are interested in assessing how changes in lending standards affect economic outcomes. However, the index of bank lending standards we construct is likely affected by economic conditions, the demand for bank lending, and other various factors (such as general perceptions of risk in financial markets), creating endogeneity issues. Our strategy involves steps aimed at addressing these issues.

Generally speaking, we need to isolate exogenous fluctuation (shocks) to lending standards that are not just reflecting the response to other forces driving bank lending, such as changes in the aggregate demand for credit in the economy. One traditional way to deal with this problem in macroeconomics is to use a VAR, which, combined with minimal structural assumptions, extracts shocks to the variables of interest. We call this the “macro approach” (pioneered in this context by Lown and Morgan (2006)). The relative simplicity of this method comes with some costs. In particular, disciplining the selection of identifying assumptions is a complicated matter with no unique correct answer ((Ramey, 2016)). In some cases, these assumptions meaningfully restrict the possibilities that can be accommodated in order to better explain the data.

More recently, Bassett et al. (2014) propose an innovative approach that combines a VAR with an initial adjustment process for the index of lending standards using bank-level data. We refer to this way of adjusting for endogeneity as the “micro approach.” An advantage of this approach is to allow the researcher to consider the possibility that exogenous changes in lending standards may affect the macroeconomy contemporaneously when specifying the macro VAR model. These shocks will be also useful to uncover some of the state-contingent effects we explore.

[7] Core lending capacity is computed as the sum of core loans outstanding and the corresponding unused commitments. We measure realized inflation as the year-over-year percentage change in core PCE. Similarly, growth in real GDP and core lending capacity are also computed as year-over-year percentage change.

In running our macro VARs, we follow the literature by imposing minimal structural assumptions and using a standard Cholesky decomposition on the residuals' variance-covariance matrix that partially restrict the contemporaneous impact of shocks.[8]

3.1. Constructing the lending standards index

The first step in our empirical strategy is to build indexes of changes in lending standards for different groups of categories and for the composite of overall bank lending activity. Taking advantage of the granular data in the construction of these indexes, we weight each bank's response by that bank's lending volume in each corresponding category (Glancy et al., 2020). This weighting approach allows us to capture more accurately the extent to which each reported change in standards affects total bank-lending activity in the economy.[9]

Based on these indexes for each category, we compute the change in lending standards for a group of categories as the weighted average of the index for each category in that group. So, for a given group c, we have:

$$\Delta S_{cit} = \sum_{k \in c} w_{c,i,t-1}(k) * I_{c,i,t}(k), \tag{1}$$

where $w_{c,i,t-1}(k)$ is the fraction of bank i's portfolio of loans in group c that is accounted for by loans in category k, with $k \in c$, according to the call reports of bank i at the end of quarter $t-1$. Weighting by category is necessary for constructing our indexes because each category has its own SLOOS questions. To construct the overall lending-standards index, we aggregate across groups of categories using the same general weighting method. Each of these indexes is a diffusion index that takes continuous values between negative 1 and 1 and can be interpreted as the *net* proportion of type-c loans in bank i's balance sheet for which the bank reported tightening its lending standards.

We then aggregate indexes across banks using the same loan-based weighting approach:

$$\Delta S_{ct} = \sum_{i \in c} w_{c,i,t-1} * \Delta S_{cit}, \tag{2}$$

where $w_{c,i,t-1}$ is the fraction of loans made by bank i of all loans in group-c categories held by SLOOS respondents at the end of quarter $t-1$. Fig. 1 plots in black ΔS_{ct} for each of the four major groups of loan categories. For each group, the plots are consistent with qualitative accounts of credit conditions during critical events in the sample period. For example, the 2007–2009 Global Financial Crisis (GFC) was preceded by a substantial tightening of lending standards for real estate, which was only later followed by tightening in other loan categories.

3.2. A micro approach to endogeneity

To address endogeneity in lending standards, the "micro approach" uses a dynamic fixed effects panel regression for the cross-section of banks in the SLOOS dataset. The regression also includes some aggregate and bank-specific factors that may be affecting loan demand. The SLOOS data includes (confidential) bank level information on the perceived evolution of demand for loans in a given quarter. The change in the assessment of loan demand conditions in group-of-categories c by each bank i at time t, denoted ΔD_{cit}, is also used as a control in the panel regression.[10]

Specifically, we regress ΔS_{cit} on a set of relevant aggregate and bank-specific variables as follows:citcit

$$\Delta S_{cit} = \beta_1 \Delta S_{ci,t-1} + \beta_2 \Delta D_{cit} + \boldsymbol{\lambda_1'} E_{t-1}[\boldsymbol{m_{t+4}} - \boldsymbol{m_t}] + \boldsymbol{\lambda_2'} * \boldsymbol{f_t} + \boldsymbol{\theta'} \boldsymbol{Z_{i,t-1}} + \phi_i + \epsilon_{cit}, \tag{3}$$

where $\boldsymbol{m_t}$ is a vector of macroeconomic variables, $\boldsymbol{f_t}$ is a vector of variables characterizing risk attitudes, and $\boldsymbol{Z_{it}}$ is a vector of bank-specific controls. Table 5 in Section 6.1 includes the regression results for the case of overall (i.e., across all categories) lending standards, where the dependent variable is ΔS_{it}.

Changes in bank lending standards tend to persist for multiple quarters as seen in the black lines in Fig. 1. By including the lagged dependent variable and bank fixed effects ($\phi_i, i = 1, \ldots, N$), we control for this cyclical momentum as well as any time-invariant factors at the individual bank level that may affect the way SLOOS respondents report changes in their credit policies. In the benchmark estimation of these micro-based adjustments we restrict the sample to exclude the COVID period. We want to avoid the possibility that the coefficients in regression (3) become overly influenced by outliers.

With the regression results on hand, we appropriately weight the estimated residuals ϵ_{cit} for each bank i to create aggregate measures of changes in bank lending standards that are purged from contemporaneous macroeconomic and bank-specific factors,

[8] Alternatively, Mumtaz et al. (2018) find, using Monte Carlo simulations, that sign restrictions can produce stable and realistic VAR-estimates when studying credit supply shocks. However, sign restrictions only lead to set identification, and these sets can be quite wide, even when robust and agnostic priors are used (Baumeister and Hamilton, 2015). Mumtaz et al. (2018)—in the spirit of Uhlig (2004) and Kilian and Murphy (2012)—also test a specification where the impact matrix maximizes the contribution of credit supply shocks to the forecast error variance. Since analyzing the relative importance of different shocks to lending standards is a key aspect of our study, we forgo this approach and take a narrower interpretation of the shocks as indicating changes in banks' willingness to make loans and not as proxies for general credit supply shocks.

[9] For each group of categories the weights are computed using the lending of each bank in each loan category. When the call-report data do not include the set of categories that would be consistent with the SLOOS data, we follow Glancy et al. (2020) in approximating the weights.

[10] We construct the bank-level demand indexes following the same steps as in the construction of the lending standards indexes.

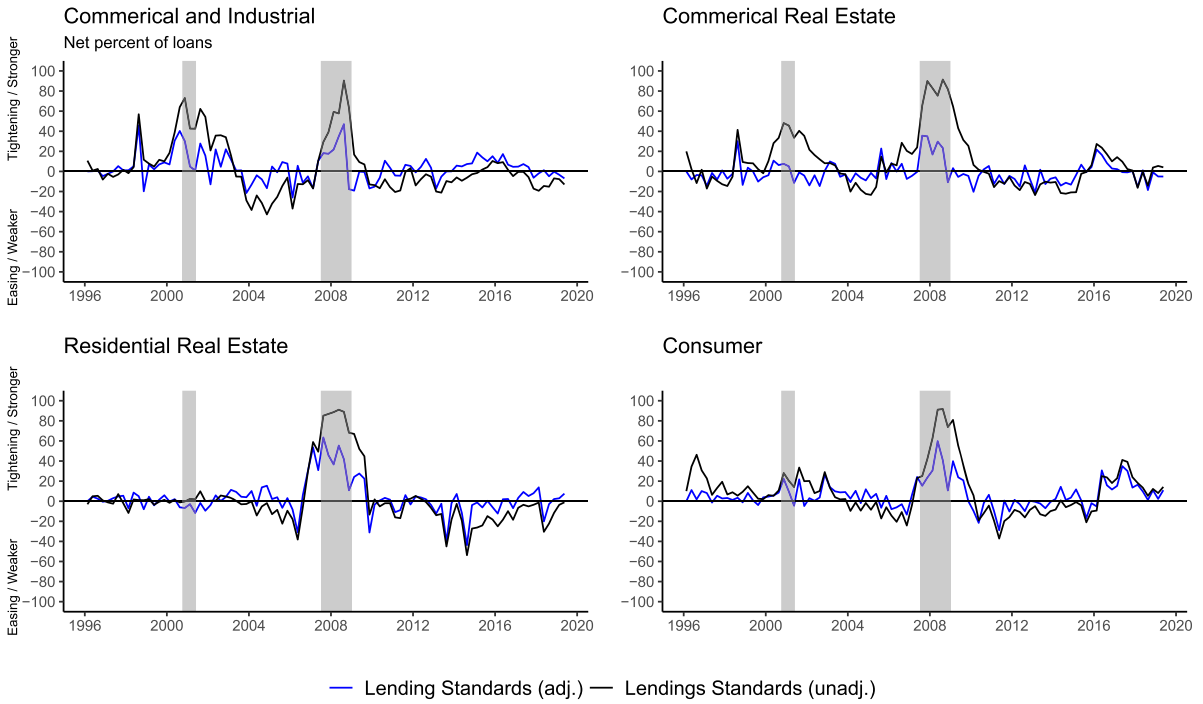


Fig. 1. Adjusted and unadjusted lending standards by group of categories
Notes: Sample period is from 1996:Q1 to 2019:Q4. Shaded areas indicate recession periods as dated by the National Bureau of Economic Research.
Sources: Federal Reserve Board Senior Loan Officer Opinion Survey on Bank Lending Practices; Consolidated Reports of Condition and Income, FFIEC 031/041/051.

including perceived (by surveyed banks) changes in loan demand. In particular, we have:

$$\Delta S^{a}_{c,t} = \sum_{i}^{N_{ct}} w_{c,i,t-1} * \hat{\epsilon}_{cit}, \tag{4}$$

where N_{ct} is the number of banks in quarter t answering questions about a loan category in group c, and $w_{c,i,t-1}$ is the ratio of loans in all categories of group c at bank i relative to the total group-c lending for the full sample of respondent banks, measured at the end of the previous quarter.

The series of *adjusted* lending standards for each broad group of loan categories, plotted in blue in the four panels of Fig. 1, suggest that credit supply tightens for consumers and businesses as the economy nears or enters a recession. For real estate, this pattern is more visible during the GFC and less evident in the 2001 recession. In turn, credit eases late in recessions or early in expansions. Interestingly, we also see that credit supply tightened noticeably early in RRE as the economy approached the GFC, consistent with the events at that time.

3.3. A macro approach to endogeneity

As we are mainly interested in the impact of changes in lending standards on the *aggregate* economy, it is reasonable to contemplate using a macro VAR estimation to directly control for some of the endogeneity in lending standards. How do results from this more traditional approach compare with the micro approach? We provide a comprehensive evaluation of this issue, which expands on the discussion in Bassett et al. (2014).[11]

To compute the index for aggregate credit demand conditions (ΔD_t), we proceed in the same way that we did to compute the aggregate index for lending standards in Eq. (2). Using this approach, we can also construct indexes for lending standards and credit demand across all groups of categories by taking a weighted average of the index for each group, using as weights the ratio of loans in each group relative to total loans in all groups. We then assess the relative performance of the micro and macro approach to

[11] Some of the macroeconomic variables used to adjust standards in the micro approach *do not* appear in macro VAR used by Bassett et al. (2014); in particular, two of these important variables not included in their VAR, the unemployment rate and the diffusion index of changes in loan demand, will be included in our macro VAR (Alternative 3) to provide a more even comparison.

Table 1
Correlations of orthogonalized residuals: alternatives 1, 2, 3 and overall adjusted standards.

	Alternative 1	Alternative 2	Alternative 3
Alternative 1	1		
Alternative 2	0.836	1	
Alternative 3	0.795	0.946	1
ΔS_t^a	0.848	0.711	0.684

Note: Sample period is from 1996:Q1 to 2019:Q4.

endogeneity by considering, in each case, three alternative specifications,

$$\mathbf{Y}_t^1 = \underbrace{\begin{bmatrix} \Delta S_t^a \\ \Delta GDP_t \\ \Delta CoreCapacity_t \\ \pi_t \\ GZSpread_t \\ RealFedFunds_t \end{bmatrix}}_{\text{Alternative 1}} \quad \mathbf{Y}_t^2 = \underbrace{\begin{bmatrix} \Delta GDP_t \\ \Delta CoreCapacity_t \\ \pi_t \\ GZSpread_t \\ RealFedFunds_t \\ \Delta S_t \end{bmatrix}}_{\text{Alternative 2}} \quad \mathbf{Y}_t^3 = \underbrace{\begin{bmatrix} \Delta GDP_t \\ Unemp_t \\ \Delta D_t \\ \Delta CoreCapacity_t \\ \pi_t \\ GZSpread_t \\ RealFedFunds_t \\ \Delta S_t \end{bmatrix}}_{\text{Alternative 3}} \tag{5}$$

Alternative 1 is the VAR that uses the micro-adjusted lending standards. Alternative 2 is the version that uses unadjusted lending standards and most closely follows the version of it in Bassett et al. (2014). Finally, Alternative 3 is a VAR that includes the variables in Alternative 2 plus two additional macro variables: unemployment and the diffusion index of changes in credit demand.

In Alternative 1, the change in adjusted lending standards appears first in the order of the VAR. The presumption is that *adjusted* lending standards have already been purged from any endogenous component, leaving only the exogenous shock to standards, which is not affected contemporaneously by any of the other variables in the VAR.[12] In alternatives 2 and 3, instead, *unadjusted* lending standards appear last in the order of the VAR. The idea here is that we intend to extract from the changes in lending standards any contemporaneous (endogenous) impact that other macro variables might have on the behavior of banks. Having unadjusted lending standards last in the order is the most likely to effectively extract all endogeneity in the process of identifying an exogenous shock to standards.

Given that the main interest here is on the effect of changes in bank lending standards on the macroeconomy, the order of the other macro variables is not critical for the results. Our ordering generally follows the previous literature but we include alternative orderings in the robustness section.

3.4. Overall lending standards and the macroeconomy

In this section, we present and compare the results from estimating the three different alternatives described in expression (5) when using as a proxy for lending standards the diffusion index aggregated across all banks and all categories. We take this as an assessment of the implications of changes in "overall" lending standards. The sample period goes from 1996:Q1 to 2019:Q4, thus excluding the COVID episode.

The lending standards shocks. Fig. 2 plots the sequence of shocks to overall lending standards extracted by using each of the three alternatives. It seems evident from the figure that the identified shocks are very similar to each other. Our expectation was that including additional variables in Alternative 3 would bring the residual closer to that from Alternative 1, since those additional variables played a role in extracting the shock in Alternative 1 but not in Alternative 2. In other words, the residual from Alternative 3 should predominantly lie between alternatives 1 and 2. While this is happening for many periods, it is not uniformly the case.

To complement the figure, Table 1 presents the correlation between the orthogonalized residuals for overall lending standards obtained using the three alternatives; what we will call, the "shock" to overall lending standards. The shock series extracted using alternatives 2 and 3 are very similar, and the ones from Alternative 2 are actually slightly more correlated with the shocks obtained in Alternative 1. The main takeaway from this table is that the extra information used in Alternative 3 does not make a considerable difference in the signal extraction process. In other words, aggregate loan demand and the unemployment rate do not seem to affect the lending standards index in a meaningful way.

Impulse response functions. We now study the impulse response functions for each of the three alternatives. Fig. 3 shows the response of macroeconomic variables to a one standard-deviation (SD) tightening shock to overall lending standards. We see that output growth falls for a year and only recovers ten to twelve quarters after the shock. All three estimation methods produce similar

[12] Ordering the exogenous shocks to lending standards first in the system is analogous to estimating the VAR as a proxy-SVAR or IV-VAR ((Plagborg-Møller et al., 2021)).

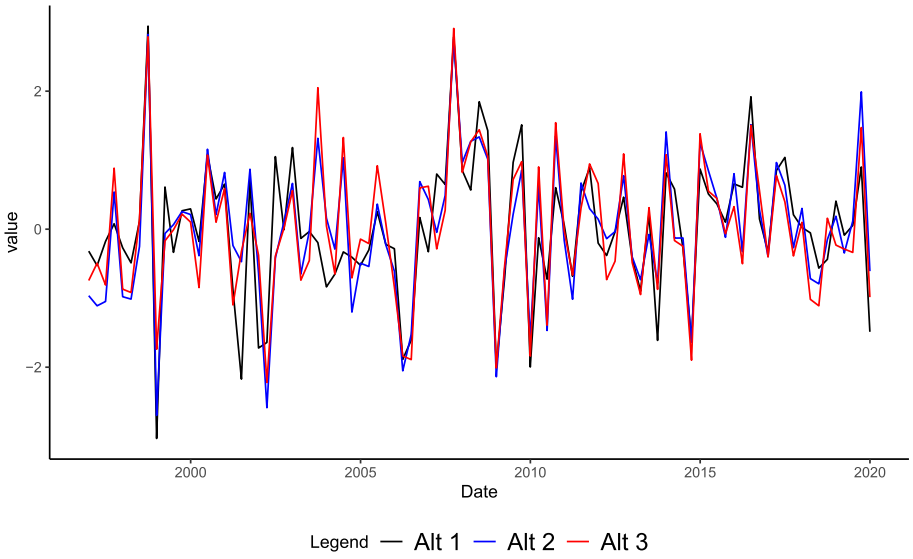


Fig. 2. Orthogonalized residuals from the alternative models
Notes: The black line corresponds to the residuals for adjusted lending standards in Alt 1 of the VARs. The blue line is the residuals for unadjusted lending standards for Alt 2 and 3, respectively. The sample period is from 1996:Q1 to 2019:Q4. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

responses, with Alternative 1 showing an immediate negative impact on output growth, which cannot be accommodated under the other two alternatives, by assumption. The impulse response for core lending capacity suggests a transmission channel for the shock, with bank lending growth falling noticeably in the quarters after the (tightening) shock.

Similarly, inflation and interest rates respond in the standard way, given the fall in output growth. But it is worth highlighting that the effect on inflation of a tightening in the supply of credit by banks is relatively subdued. Corporate bond spreads, on the other hand, widen noticeably as banks tighten credit. The inclusion of core bank lending capacity makes virtually no difference for the estimation of the impulse responses of output, inflation, and the interest rate. Including it, however, allows us to gain some insight into the mechanism propagating the shocks.

The bottom panel of the figure shows that fluctuations in lending standards display considerable persistence (with shocks dissipating only after six or seven quarters—blue and red lines in the figure). It is also useful to see that our measure of shocks to lending standards obtained with the micro approach (the black line in the figures) has, as expected, little-to-no persistence, which suggests that the adjustment does a good job in isolating the shock component of lending standards.

With the estimation of Alternative 3, we also obtain an impulse response for unemployment and loan demand. As expected, unemployment increases as output growth falls. Loan demand (as perceived by banks) falls after the shock to standards, but the response is relatively mild. Interestingly, loan demand actually recovers and becomes positive after ten quarters, when the effect on output of the shock is almost fully dissipated.

It is also clear from Fig. 3 that all three alternatives produce very similar impulse response functions. Aside from the contemporaneous effect on output estimated under Alternative 1 (which is not statistically significant), the rest of the effects are basically the same across all alternatives. One interpretation of these results is that the micro adjustment does not produce significantly different outcomes when it comes to understanding the macroeconomic effects of changes in lending standards. Given that for macroeconomic purposes we aggregate lending standards (adjusted or not), the process of aggregation tends to minimizes the role of the idiosyncratic cross-sectional variation removed by the micro approach. As a result, the series of *aggregated* lending standards, adjusted or unadjusted, capture very similar macroeconomic forces.

4. Lending standards for businesses and households

We now seek to get a deeper understanding of the impact of shocks to lending standards on the macroeconomy. In principle, we could proceed by estimating the effect of changes in lending standards separately for each of the four main groups of loan categories. However, Table 2 suggests a possible further grouping of these categories. Given that C&I and CRE are highly correlated and the correlation of consumer loans and RRE is relatively high as well, particularly for the micro-adjusted data (second panel in the table), we adopt two supra-groups: lending to business (C&I and CRE) and to households (CONS and RRE).

The correlation of unadjusted lending standards among the two real estate groups of categories is also relatively high, but falls considerably for the adjusted series. This suggests that the high correlation in standards across real estate categories is mainly driven by endogenous factors (the response of standards to common macroeconomic factors). Since all our estimation alternatives intend to

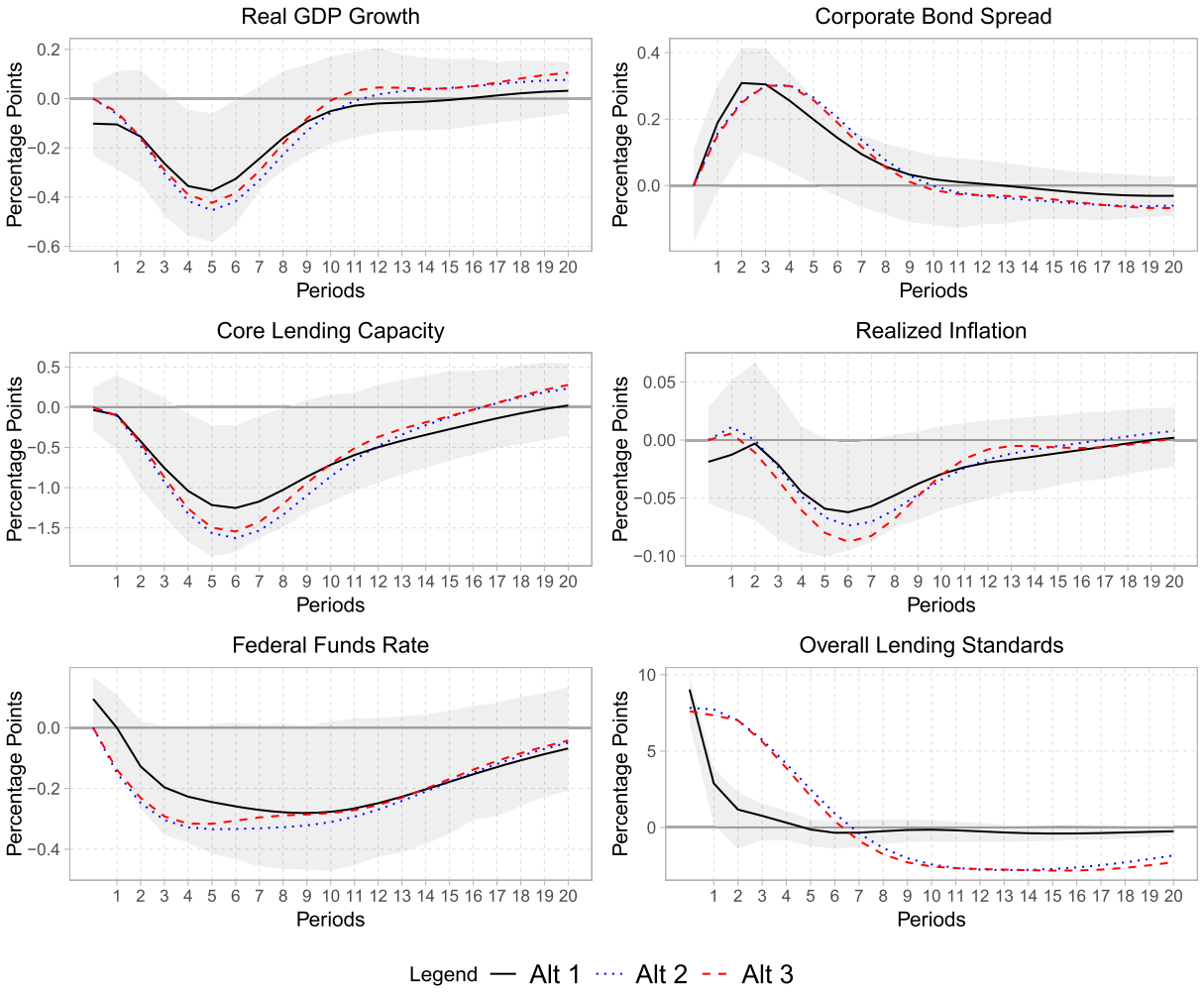


Fig. 3. Orthogonal impulse response functions: Shock to overall lending standards
Notes: The shaded area represents bootstrap 95% confidence intervals from 250 iterations, centered on Alternative 1 (in black). Alternative 1 uses a 1 SD shock to ΔS_t^a, and alternatives 2 and 3 use a 1 SD to ΔS_t. All models are estimated using 2 lags. The sample period is 1996:Q2 to 2019:Q4.

Table 2
Correlations of lending standards between groups of categories.

	Unadjusted (ΔS_{ct})					Adjusted (ΔS_{ct}^a)			
	C&I	RRE	CRE	CONS		C&I	RRE	CRE	CONS
C&I	1				C&I	1			
RRE	0.522	1			RRE	0.167	1		
CRE	0.781	0.762	1		CRE	0.590	0.373	1	
CONS	0.600	0.689	0.695	1	CONS	0.331	0.449	0.291	1

Note: Sample period is from 1996:Q1 to 2019:Q4.

control for endogeneity, the relevant correlation to consider for grouping categories is the one that abstracts from this endogeneity. Our choice of supra-groups of categories is consistent with this logic.

In the robustness section, we include an analysis of each of the four groups of categories separately. That analysis, in fact, provides further support for grouping categories the way we do here, which translates into a valuable gain of parsimony without any significant loss. Glancy et al. (2020) also adopt this mode of aggregation to summarize the data.

To adjust lending standards at the supra-groups level, we run a separate panel regression for each of the two supra-groups, following the same approach as in expression (3). Fig. 4 plots the time series of the indexes for unadjusted and adjusted lending standards. Standards tighten during recessions, and it is clear from the figure that during those times, endogenous amplification plays

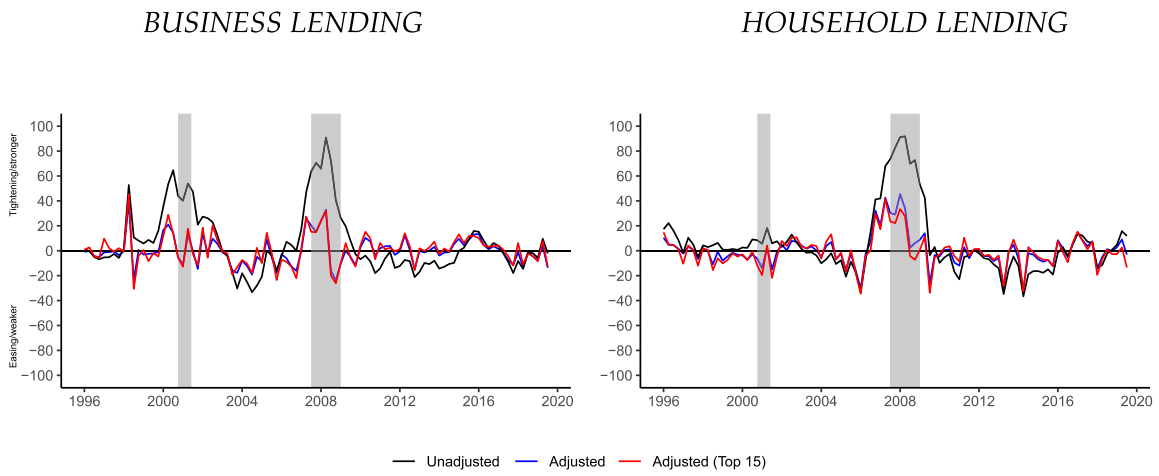


Fig. 4. Unadjusted and adjusted lending standards for supra-groups
Notes: The plot on the left is for standards in business lending (BUS), and the right panel is for standards in lending to households (HOUS). In each figure, the black and blue lines represent standards for the FULL sample of SLOOS banks. The red line shows adjusted standards for the top 15 SLOOS banks in our sample, for future reference (see Section 6.2). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

a significant role driving bank lending standards. Lending standards for both businesses and households tightened in comparable ways during the GFC. The effect of the 2001 recession appears particularly strong for business lending standards.

4.1. Business and household lending standards and the macroeconomy

We estimate three VARs corresponding to each of the three alternative approaches introduced in expression (5), but we now include two separate indexes for lending standards: one for business lending and one for household lending. Having two components of lending standards in the VAR brings up a new issue: which of the two components is to appear first in the ordering of the variables? In our benchmark specification we have business standards coming right before household standards in the VAR order, but results are effectively unchanged when that order is reversed.[13]

Fig. 5 plots the impulse responses functions. We include the main macroeconomic variables: output, inflation, and the interest rate and consider, side by side, a one standard deviation tightening shock to lending standards for business (on the left) and for households (on the right). It is clear from the plots that the shock to business lending has the more significant effect over output and other macroeconomic variables. In fact, the bootstrap confidence intervals indicate that the effects of a shock to household lending standards are not statistically significant at all horizons.

These impulse response functions suggest that the effects we observed in Fig. 3 are mainly driven by the tightening of business credit, with the tightening of credit to household only reinforcing those effects. Confirming our prior assessment, we see in Fig. 6 that lending to businesses drops significantly after the shock, driving in large part the macroeconomic impact. Lending to households, on the other hand, responds much less to a shock on bank credit to households.

In principle, multiple reasons can be behind this disparity between business and household credit. First, banks are a more critical source of funding for businesses than for households. Furthermore, it is well known that investment is more cyclical and volatile, and the impact of standards on investment could be more significant for that reason. The findings in Cavallo et al. (2024a) support this interpretation. They study how a shock to bank credit affects economic activity measured by private domestic final purchases. This allows them to decompose the aggregate effects. They find that investment in equipment and structures are the demand components that are most responsive to shocks in lending standards.

Another factor that might be contributing to this observed disparity in measured effects is that the provision of unsecured credit to households tends to be more concentrated (in a few credit-card banks, for example) and less susceptible to exogenous changes in lending standards (such as the ones analyzed here) because the approval process is based on explicit metrics and is highly automated.

A final consideration is that lending standards for households are more difficult to aggregate accurately. The match between SLOOS and call-report categories is less exact in the case of household lending. Furthermore, residential mortgages are regularly packaged and sold, which implies that banks' holdings of residential mortgages are a rather imperfect proxy for the influence in the economy of bank lending standards tied to the origination and distribution of those loans.

[13] Going back to expression (5), for all three alternatives, we replace ΔS by ΔS^{BUS} *and* ΔS^{HOUS} (both) in those specifications, maintaining the general ordering of variables.

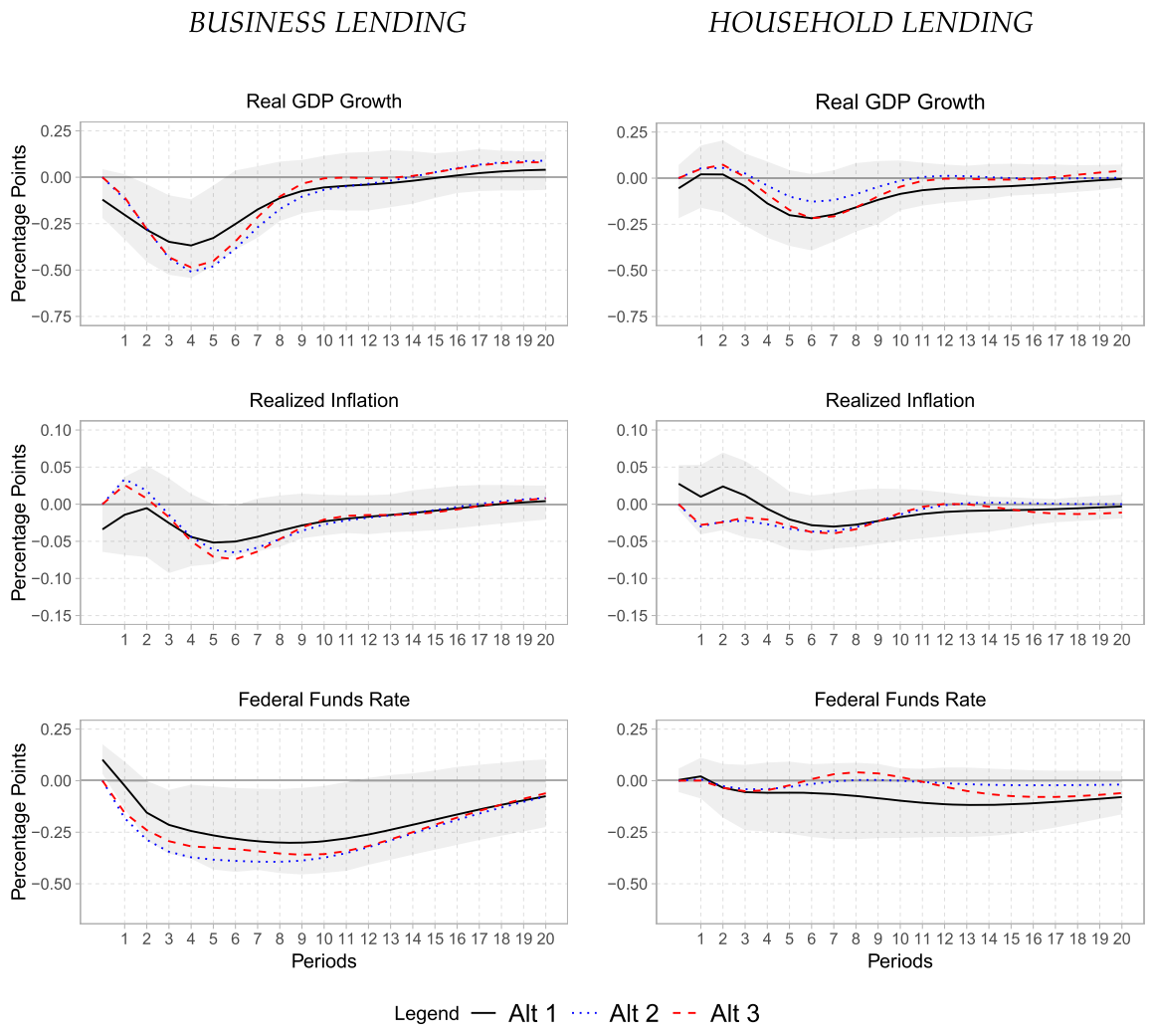


Fig. 5. Orthogonal impulse response functions: Shock to business and household lending standards
Notes: The left column corresponds to the effect of a 1 SD shock to business lending standards and the right column a 1 SD shock to household lending standards. The shaded area represents bootstrap 95% confidence intervals from 250 iterations, centered on alternative 1 (in black). All models use 2 lags. The sample period is 1996:Q2 to 2019:Q4.

While these measurement issues are worthy of consideration, we do not think they are the main cause of the disparity we uncover. The variance decomposition we discuss next indicates that lending standards for households have some explanatory power at medium-term horizons. Similarly, the residuals in business and household lending standards have similar-sized variance, which suggests that random noise is not more predominant in household standards relative to business.

4.2. The relative importance of bank lending standards shocks

An important and related question to consider is how much of the variability in the main macroeconomic variables included in our VARs can be attributed to the different shocks at different forecast horizons, and in particular, to which extent do lending standards drive this variability. To this end, Table 3 presents the forecast error variance decomposition for real GDP growth, inflation, and the fed funds rate of a VAR estimated using *adjusted* lending standards for business and household (Alternative 1).

The table provides the decomposition at different horizons indicated by h in the second column. The entries on the table represent the percentage contribution of each of the shocks to the h-quarters-ahead forecast error variance for each of the three main macro variables. The entries in each row sum to 100 percent (after rounding). In most cases, after ten quarters, the variance decomposition settles down and becomes a good approximation of the unconditional variance decomposition. Shocks to business and household lending standards account for one quarter of the variance in real GDP growth after ten quarters. Most of the effects come from changes to business lending standards. At shorter horizons, household lending standards contribute minimally to explain fluctuations

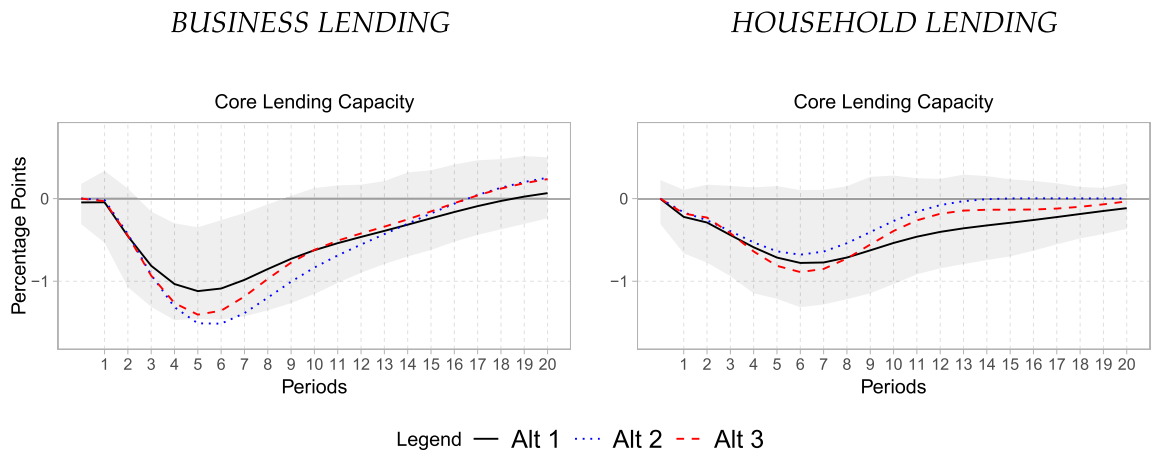


Fig. 6. Impact on bank core lending capacity of a shock to business/household lending standards
Notes: The left panel corresponds to the effect of a 1 SD shock to business lending standards and the right panel a 1 SD shock to household lending standards. The shaded area represents bootstrap 95% confidence intervals from 250 iterations, centered on alternative 1 (in black). All models use 2 lags. The sample period is 1996:Q2 to 2019:Q4.

Table 3
Forecast error variance decomposition for Alternative 1 of the VAR.

Forecast error in	Forecast horizon	Percentage of variance, *h* periods ahead, accounted for by innovations in						
	h	BUS	HOU	RGDP	CC	INF	CBS	FFR
RGDP	1	7.5	0.5	85.7	4.2	1.3	0.9	0.0
	5	19.4	2.5	37.2	3.5	29.6	7.6	0.1
	10	20.0	6.4	32.2	3.7	30.4	7.1	0.3
	15	19.4	6.6	31.1	3.8	31.0	6.9	1.2
INF	1	2.0	1.3	1.3	2.6	90.8	1.6	0.5
	5	5.2	1.6	0.9	9.7	69.6	8.1	5.1
	10	9.3	3.6	1.2	8.5	61.2	8.2	8.0
	15	9.8	3.9	1.5	8.4	60.2	8.2	7.9
FFR	1	4.0	0.2	12.1	0.2	24.9	0.2	58.4
	5	14.1	0.8	16.5	2.8	10.7	0.1	54.9
	10	21.5	1.4	21.2	1.9	11.0	0.5	42.6
	15	24.1	2.8	21.3	1.5	14.1	1.1	35.0

Notes: Forecast error variance decomposition for real GDP, realized inflation, and the federal funds rate. The variance decomposition is constructed via Cholesky decomposition, using quarterly data from 1996:Q2 to 2019:Q4. All variances are reported as percent of the total error variance at horizon h attributed to each given shock.

in aggregate output. Interestingly, lending standards to businesses appear to play a significant role in explaining fluctuations of the fed funds rate, even if the effects on inflation are relatively moderate.

Alternative 1 uses lending standards that are adjusted with micro and macroeconomic data and then appear first in the variables' order of the VAR. This specification is not well-suited to think about feedback effects from macroeconomic conditions onto lending standards; i.e., how much of the changes in lending standards are the response to changes in real GDP growth and, more broadly, the general macroeconomic outlook. To get a perspective on this issue, we use our estimation based on Alternative 3 to compute the forecast error variance decomposition of bank lending standards for both businesses and households. The results appear in Table 4.[14]

We see in the table that real GDP growth explains a significant portion of the fluctuations on lending standards, particularly for businesses. For households, this effect of output over standards is present, but less strong. Business lending standards, on the other hand, seem to affect household lending standards considerably after a few quarters (second to last column of the bottom panel). On the demand side, household demand for bank lending appears to be a more important driver of lending standards to households, in the medium term, relative to the role of business demand for lending standards to business, even though at shorter horizons, business lending standards appear more responsive to changes in demand shocks.

[14] The online supplementary material includes a more complete version of these tables.

Table 4
Forecast error variance decomposition for Alternative 3 of the VAR.

Forecast error in	Forecast horizon	Proportions of forecast error variance h periods ahead accounted for by innovations in									
	h	RGDP	Unemp.	BUS_d	$HOUS_d$	CC	INF	CBS	FFR	BUS_s	$HOUS_s$
BUS_s	1	13.2	3.1	8.9	0.9	2.8	0.9	8.7	3.3	56.5	1.6
	5	11.2	7.1	5.8	2.5	3.2	0.6	9.3	3.0	52.3	4.9
	10	12.1	6.2	5.6	6.0	4.6	0.7	8.8	4.5	46.9	4.7
	15	15.0	5.2	5.8	5.3	4.0	0.8	9.4	4.0	46.6	4.0
HOU_s	1	4.1	1.9	0.5	0.8	0.3	2.8	1.7	3.5	9.2	75.1
	5	6.8	6.1	0.8	9.9	0.3	9.1	1.4	1.7	15.5	48.4
	10	5.9	5.4	0.7	19.1	0.5	13.0	2.6	1.6	13.8	37.4
	15	8.5	4.6	1.2	18.9	0.5	11.7	3.5	1.5	17.6	32.1

Notes: Forecast error variance decompositions for business (BUS) and household (HOUS) bank lending standards. The variance decompositions are constructed via Cholesky decomposition, using quarterly data from 1996:Q2 to 2019:Q4. All variances are reported as percent of the total error variance at horizon h attributed to each given shock.

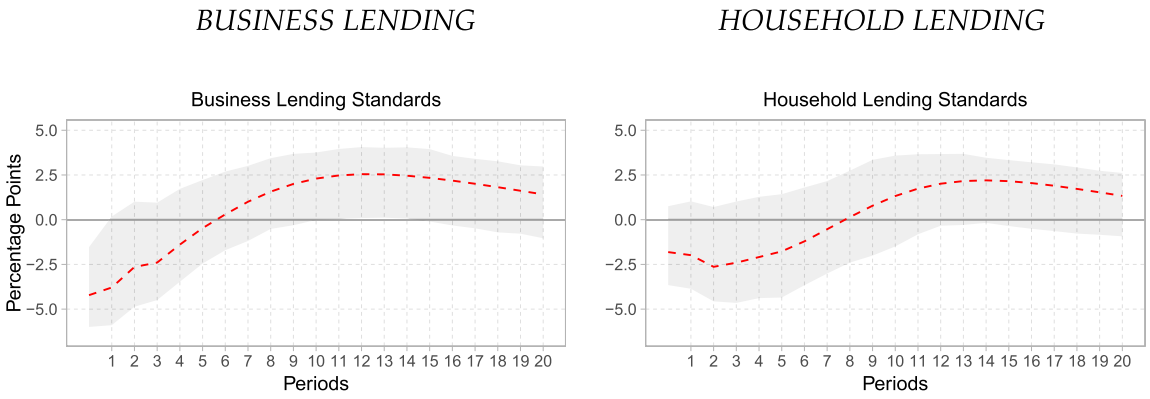


Fig. 7. Impact on lending standards of a shock that impacts real GDP growth
Notes: The left panel corresponds to the effect of a 1 SD shock to output growth on business lending standards, and the right column is the effect on household lending standards of the same shock. The shaded area represents bootstrap 95% confidence intervals from 250 iterations. The VAR includes 2 lags. The sample period is 1996:Q2 to 2019:Q4.

To complement this information, Fig. 7 plots the impulse response of lending standards to the shock affecting output growth —the first variable in the order— in the estimated VAR used to produced Table 4 (Alternative 3). We see that a shock increasing output growth tends to initially loosen business lending standards, with a reversal after seven to ten quarters. The effect of GDP shocks on household lending standards shows a similar pattern, but much smaller and mostly insignificant.

In general, these results point to the existence of sizable feedback effects: a tightening in business lending standards tends to drive a slowdown in real GDP growth after a few quarters, which in turn induces further tightening of those standards, reinforcing the initial shock.

5. Asymmetric and state-dependent effects

In this section, we look into the nature of the effects that changes in bank lending standards have on the macroeconomy. We consider two possibilities: (1) that the effects are different depending on the state of the economy; and (2) that the effects are asymmetric, with positive shocks affecting the economy differently than negative shocks (Barnichon et al., 2022).

5.1. Empirical framework

The basic approach is based on the VAR we used before, although we now allow the matrix of VAR coefficients to be contingent on the state of the economy. In particular, we adopt a threshold VAR specification, or TVAR, similar to that in Section 18.3 of Kilian and Lütkepohl (2017).

Denote by Θ the matrix of coefficients for a given VAR specification, and let S be the set of mutually exclusive and predetermined economic states s under consideration. At each time t, the economy will be in one of these states—more formally, for each time t in the sample period, the state of the economy $s(t)$ equals s for some $s \in S$. In the estimation, we let the matrix Θ change when the state changes; hence we write Θ_s. For this methodology to be operational in our context, the number of elements of S has to be relatively small to have sufficient data under each possible state.

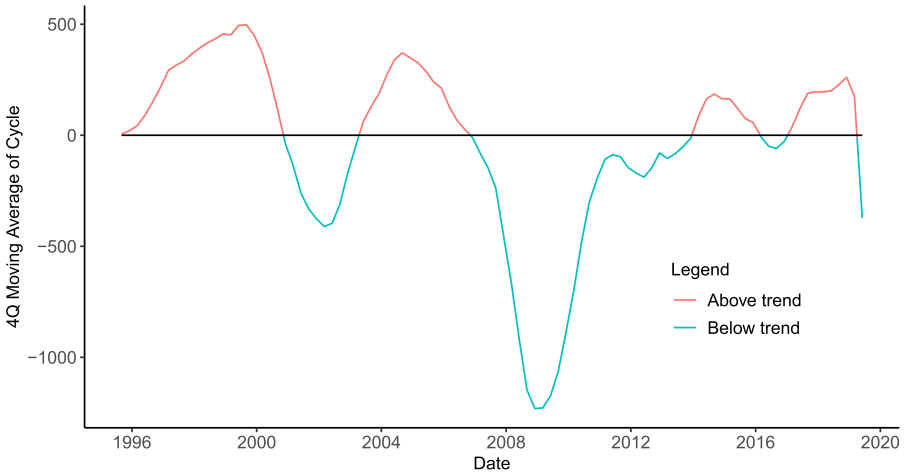


Fig. 8. Real GDP growth, above and below trend split
Notes: We take quarterly real GDP (billions of chained 2012 dollars, seasonally adjusted annual rate) and decompose the trend and the cycle using Hamilton's filter. We then take the 4-quarter moving average of the cycle component (averaging the prior quarter, the current quarter, and the next two quarters). When this moving average is above 0 we say that GDP is above trend, which we indicate with red color. Blue, on the other hand, indicates quarters we classified as having output growth being below trend. The sample period is 1996:Q2 to 2019:Q4. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

To determine the economy's state at each point in time, we use relatively simple criteria based on patterns of the data that are not part of our VAR specification. Given the relatively short sample period, we restrict the number of states in the TVAR to one or two and use a parsimonious lag structure to produce appropriate identification.[15]

The generalized, state-contingent VAR can be written concisely as:

$$Y_t = \Theta_{s(t-1)} Y_{t-1} + \Sigma_{s(t-1)} u_t, \quad u_t \sim \mathcal{N}(\mathbf{0}, I), \tag{6}$$

where, for all t in the sample period, $s(t-1) = s$ for some $s \in S$, and the model allows the joint distribution of shocks Σ to be influenced by the state, as well. Persistence and propagation are, of course, also influenced by the state via the state-contingent matrix Θ_s. Note that the model encompasses the non-state-contingent VAR and, if the data generating process is not meaningfully state-dependent, then the estimation will show no significant differences across states.

The main objective here is to enhance our understanding of the economic mechanisms driving the effects of changes in lending standards on the macroeconomy. This will motivate the choice of the set of states S and is also the reason why we proceed with a simple (and easily interpretable) TVAR model, instead of using a smooth-transition or a regime switching VAR. Similarly, when we plot impulse responses, we will assume that the economy does not switch regimes in the subsequent periods after the shock. For the purpose of forecasting, it would be more appropriate to consider generalized impulse response functions, which are more demanding in terms of the amount of data needed to obtain acceptable identification.

The exercises in this section should be interpreted as exploratory of the forces behind the macro effects identified in Fig. 3. A full characterization of the state-dependent dynamics is not possible with our limited data. When we split the sample by states, each set of state-contingent parameters is identified using only part of the data, and hence less information. This is also why we analyze alternative state-contingencies *one at a time*. As will become clear, the states associated with the different contingencies produce subsamples that overlap. In the future, as more data becomes available, it would be interesting to create a partition with multiple states that account for these various contingencies *simultaneously*.

5.2. The state of the business cycle

In this section, we consider the case when the set S has two elements capturing the state of the business cycle. For our baseline estimation, we use the *level* of output, relative to trend, as the criterion for defining the states. That is, one state relates to periods when output is above its long-run trend, and the other when output is below trend. We use Hamilton's method to determine the trend in GDP (Hamilton, 2018) and we say that output is above (below) trend when the four-quarter moving average of the detrended series is positive (negative) for at least two consecutive quarters. For our sample period, there are 51 quarters with output above trend and 44 quarters with output below trend. Fig. 8 shows the split of our sample period according to these two states.

[15] In most case, this requires just one lag, which is also the BIC-minimizing number in all cases.

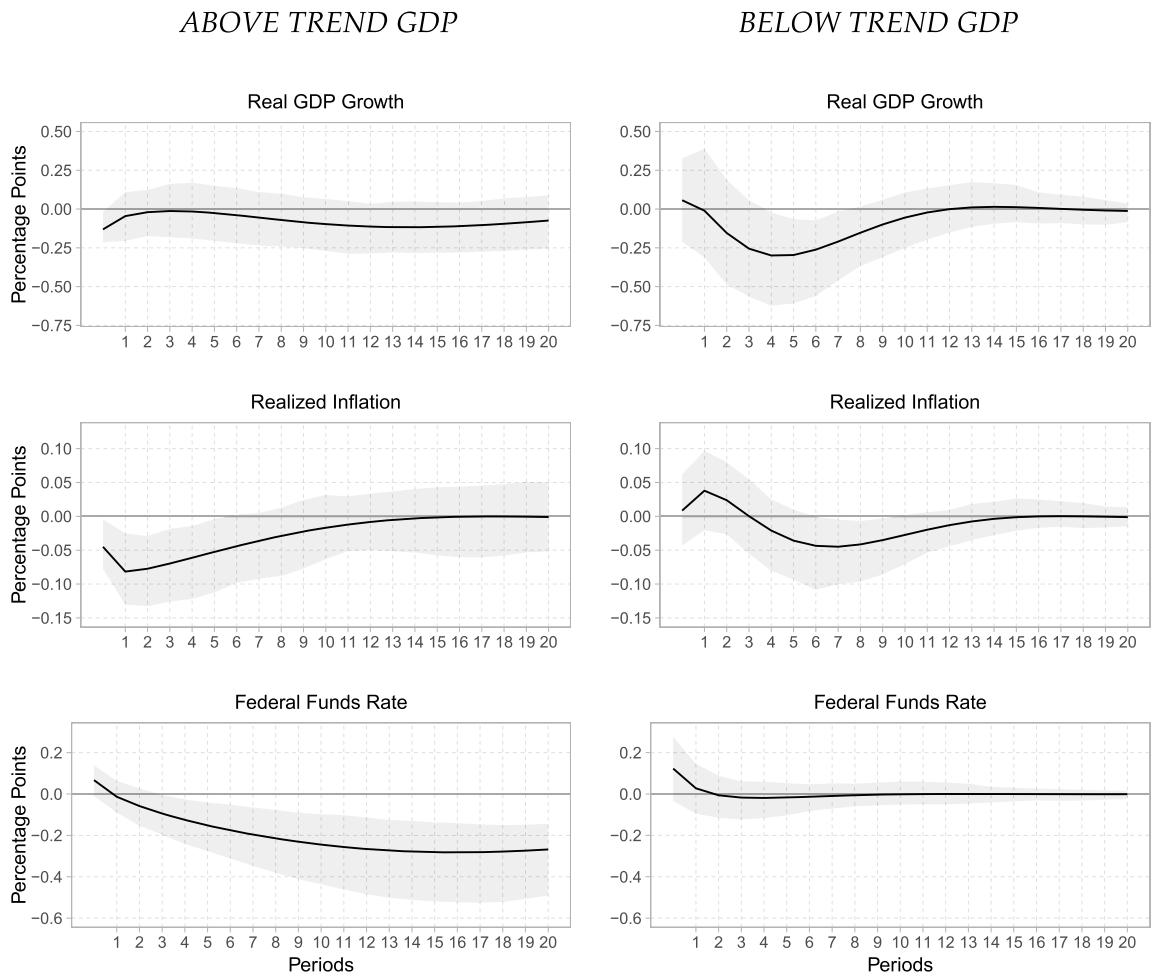


Fig. 9. State of the business cycle: OIRF for a shock to overall lending standards
Notes: Based on the estimation of Alternative 1, the plots correspond to the effect of a 1 SD tightening shock to lending standards that happens when real GDP is above trend (left column) and below trend (right column). The shaded area represents bootstrap 95% confidence intervals from 250 iterations. All models use 1 lags. The sample period is 1996:Q2 to 2019:Q4.

Fig. 9 plots the resulting state-contingent impulse response functions. It is clear from the figures that most of the impact on output growth from shocks to lending standards comes from times when output is below trend. The effects on inflation and interest rates are much more subdued.

We also verify, as before, that the effect on GDP growth of a tightening in standards when output is below trend occurs in conjunction with a contraction in banks' core lending capacity. This again suggests a mechanism where banks tighten standards, credit falls, and, consequently, output reflects the fact that bank loans become harder to access.

When output is above trend, our impulse response functions indicate that adjustments in banks' lending standards do not have much impact on GDP growth. This asymmetry between the effects of a tightening of lending standards during expansions and contractions in economic activity is consistent with the traditional financial accelerator literature (Bemanke and Gertler, 1989). During contractions, the financial net worth of productive firms suffers, and this increases the external finance premium, making external funding more expensive. Banks' tightening of lending standards forces customers to look for credit elsewhere at a time when external financing is particularly expensive. On the other hand, a tightening of bank credit during an expansion is less likely to significantly affect access to credit by productive firms, and non-bank sources of funding can quickly fill the void.

Another interesting pattern in Fig. 9 is that a tightening in lending standards tends to have a significant (negative) effect on inflation when output is above trend. And, in that case, the fed funds rate eventually accommodates the tightening shock with a persistent decrease that lasts for several years.

More generally, this impulse response function suggests that the ability of interest rates to respond to a tightening of lending standards can play a role in how lending standards impact GDP growth. With this in mind, we now turn our attention to the case when we allow parameters in the VAR to depend on the level of interest rates.

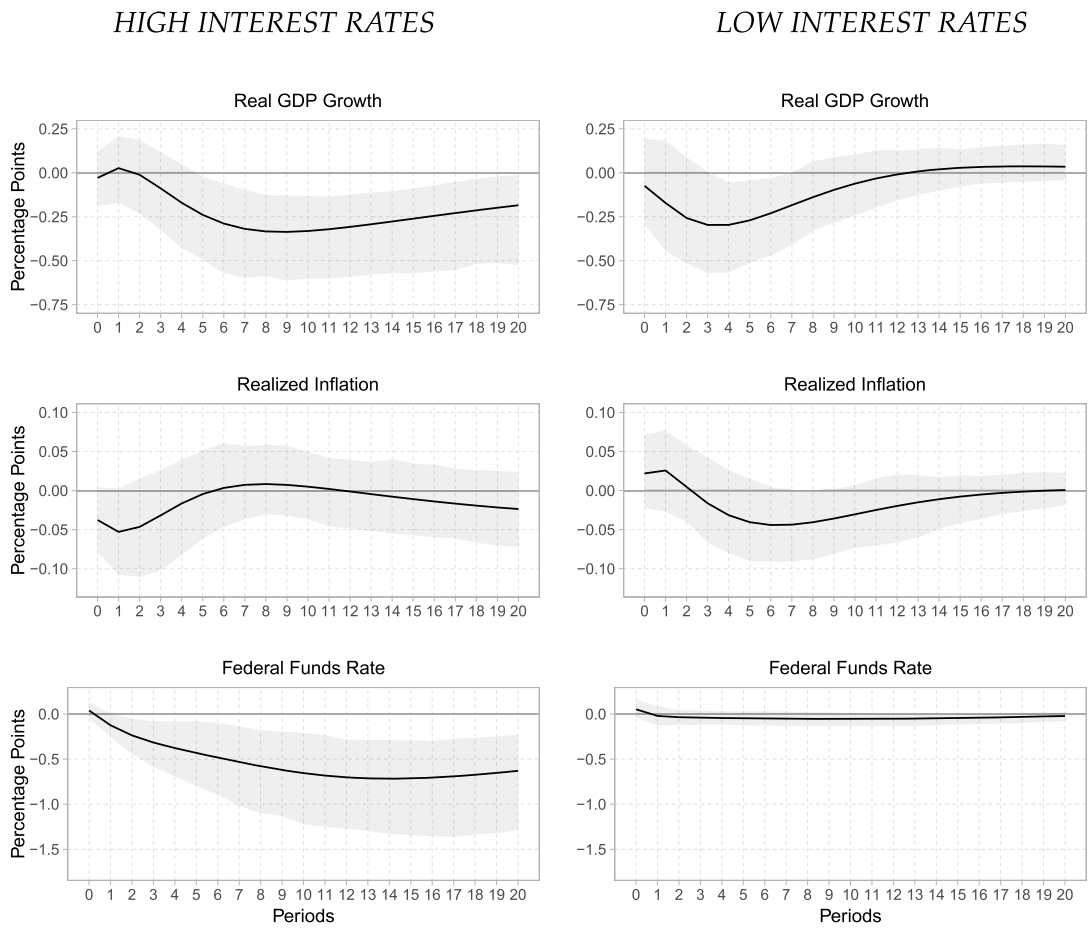


Fig. 10. Proximity to the ZLB: OIRF for a shock to overall lending standards
Notes: Based on the estimation of Alternative 1, the plots correspond to the effect of a 1 SD shock to lending standards that happens when the nominal fed funds rate is below 1.5% (left column) and above 1.5% (right column). The shaded area represents bootstrap 95% confidence intervals from 250 iterations. We use 2 lags in the estimation. The sample period is 1996:Q2 to 2019:Q4.

5.3. Proximity to the effective lower bound on interest rates

We now consider the case when, again, S has two elements, according to the proximity of the level of the nominal interest rate to its effective lower bound (ELB). The idea is to assess whether shocks to lending standards affect the economy differently when there is a greater probability of having the lower bound on interest rates bind.

We set the threshold for the nominal fed funds rate to 2 percent and designate the low interest rate state as the one including all quarters when the rate is below that threshold. We have 54 low-interest rate quarters and 41 high-interest rate periods. We select the 2-percent threshold partly in order to balance the informational content in the resulting split of the data.

Fig. 10 plots the impulse response functions for the main macro variables in response to a tightening shock to lending standards. The effect on output growth of a tightening of lending standards is significant and very persistent during high-rate periods. Interest rates fall and inflation falls initially but mostly remains stable in response to the shock. During low interest rate periods, on the other hand, the effects on output growth are not as persistent and the effect on inflation is insignificant. Naturally, the fed funds rate appears highly insensitive to a shock in the low-interest-rate state, as there is not much room to accommodate the impact by lowering rates in such situations.

There are several reasons discussed in the literature for why proximity to the ZLB may dampen the response of the economy to financial shocks (see, for example, Abadi and Koby (2023) and Eggertsson et al. (2024)). According to this work, when the economy is operating in a persistently low interest-rate environment, bank credit and risk taking ebb, with interest margins compressed and reduced internal capital generation. Based on this reasoning, a tightening shock to lending standards can have a more subdued impact on bank activity and general economic conditions, as banks already operate in a conservative mode and there is limited room to further contract credit in response to the tightening shock.

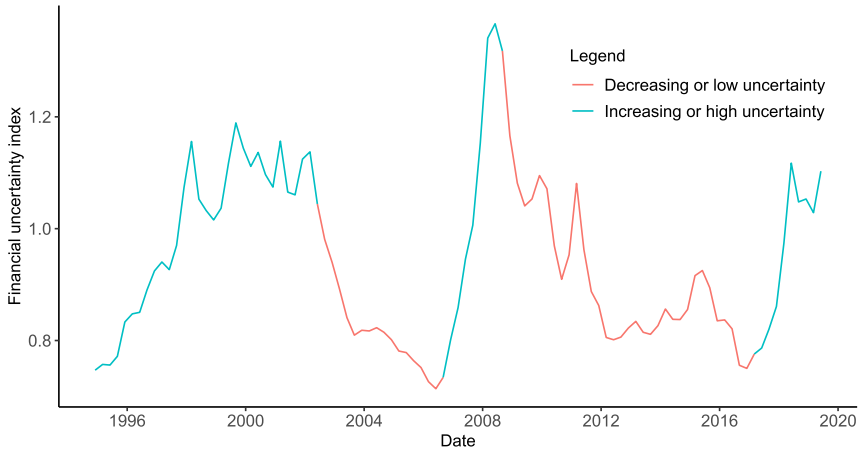


Fig. 11. Increasing or high vs decreasing or low uncertainty
Notes: The red color indicates quarters when the index of financial uncertainty is decreasing or stays low after decreasing consistently, and the blue indicates periods when uncertainty is increasing or stays high after increasing consistently. The sample period is from 1996:Q1 to 2019:Q4. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

5.4. Periods of increasing-or-high and decreasing-or-low uncertainty

Forni et al. (2024) study the role of nonlinearities in driving the impact of financial shocks on the economy. One aspect that they investigate in detail is the possibility of state-dependent effects, like we do. They allow effects to be conditional on the state of the business cycle, the level in interest rates, and other various factors. Interestingly, the only state variable that seems to make a meaningful difference in their study is the level of macroeconomic uncertainty: they find that the effects of a financial shock are stronger during periods of high uncertainty.

There are several differences between our specification and that in Forni et al. (2024). Most importantly, we are focusing here on understanding a shock to lending standards, while Forni et al. (2024) consider a more general financial shock using a longer sample period and monthly data. In light of their results, and these differences notwithstanding, we investigate here how our results depend on the evolution of financial uncertainty.

We partition the sample into two states according to the index of financial uncertainty introduced by Ludvigson and Serena (2021): one state corresponds to periods when uncertainty is decreasing and eventually stabilizes at a low level; the other state is when uncertainty is increasing and eventually stabilizes at a high level. Fig. 11 shows the split of our sample period according to these criteria. We have three periods of increasing-or-high uncertainty (in blue) and two periods of decreasing-or-low uncertainty (in red).

Fig. 12 plots the impulse response of real GDP growth to a tightening shock to lending standards in each regime. Much of the unconditional effects we identify in Fig. 3 come from periods of increasing-or-high uncertainty: A tightening shock to overall lending standards during those periods tends to reduce real GDP growth in a persistent manner.[16]

Similar to the other cases, we verify that core lending capacity drops significantly in response to a tightening of lending standards in periods of increasing uncertainty. A possible explanation is that, as uncertainty increases, it becomes harder for bank customers to find alternative sources of funding outside the banking system. Greater uncertainty also induces agents to become more conservative, which can amplify the response to a negative credit shock, as in Buera and Nicolini (2020). This kind of amplification effect is most relevant when there is a negative (rather than a positive) credit shock. We consider such asymmetries next.

5.5. Asymmetric effects of tightening and easing shocks

Here, we estimate a version of the model which allows the VAR parameters to be contingent on whether the adjusted index of changes in overall lending standards is positive or negative. More specifically, S has two elements, one denoting the state when exogenous changes to lending standards are positive and one when they are negative. As before, we allow for changes in the state only when the shock is of the same sign for at least two consecutive periods (to reduce high-frequency switches that are unlikely to represent meaningful economic effects).

Notice that for this exercise, it is critical to have the *adjusted* lending standards, which are a representation of exogenous shocks. This allows us to directly classify changes as positive or negative shocks, and to use that classification of regimes to then estimate

[16] The protracted drop in economic growth eventually gets reflected in lower inflation rates. The real interest rate also falls, consistent with our previous findings. We use 2 lags in the estimation of this version of the TVAR, as it allows the model to better handle stationarity of the estimated process.

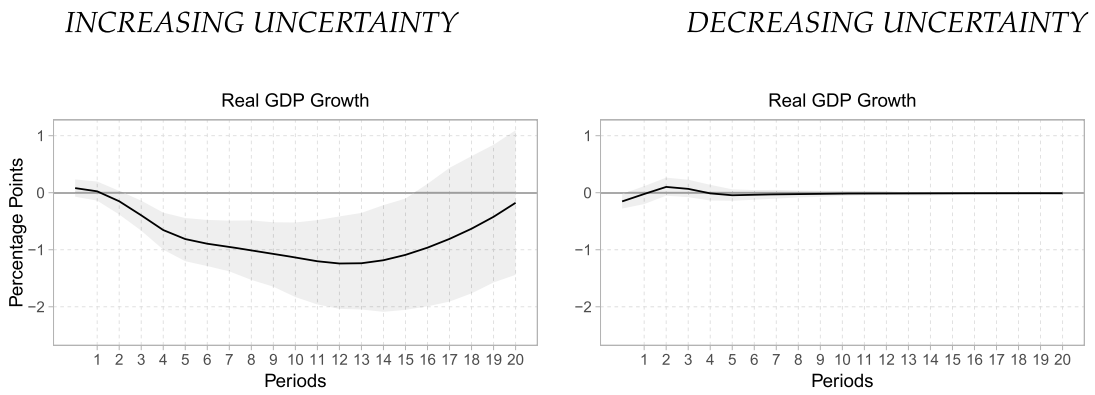


Fig. 12. Financial uncertainty: OIRF for a shock to overall lending standards
Notes: Based on the estimation of Alternative 1, the plots correspond to the effect of a 1 SD shock to lending standards that happens when the index of financial uncertainty is increasing (left column) and decreasing (right column). The shaded area represents bootstrap 95% confidence intervals from 250 iterations. We use 2 lags in the estimation. The sample period is 1996:Q2 to 2019:Q4.

state-contingent parameters for the VAR. In contrast, the VARs in alternatives 2 and 3 of expression (5) use the unadjusted lending standards and obtain the sequence of shocks as a by-product of the VAR estimation. Conditioning on the sign of the shock is much more complicated in that case (Barnichon et al., 2022).

Fig. 13 plots the impulse responses of the main macroeconomic variables to a one-standard-deviation shock to overall lending standards. In the left panel, we plot the response to a positive (tightening) shock to lending standards, and in the right panel, we plot the responses to a negative (easing) shock to standards. A tightening shock has a negative and significant impact on GDP growth. Easing shocks to standards, on the other hand, have a more subdued effect on output growth, mostly statistically insignificant, but a gradual and long-lasting positive effect on inflation and interest rates.

Fig. 14 shows that the movement in GDP growth is also consistent with the effect that standards have on banks' core lending capacity. In particular, a tightening shock to lending standards tends to lower bank lending, consistent with the transmission of the shock going from bank lending to output. This is in line with the findings in Castro et al. (2022b) who show that the response of banks to idiosyncratic credit supply shocks is asymmetric in the same direction as we find here.

Forni et al. (2024) study whether the *size* of the shock is also important in accounting for the effects of financial shocks on the economy. In particular, they find that only sufficiently large shocks have a meaningful effect. In our context, since we find that the *direction* of the shock is important, one would want to condition simultaneously on size and direction. We are not able to do this due to data limitations: there are only a handful of large tightening shocks in our sample period. [17]

6. Robustness

In this section, we conduct several robustness exercises that reinforce and expand our results. Given the findings so far and the nature of the exercises we do here, we focus mostly on overall lending standards as the leading example. Similar results can be obtained with more disaggregated series for lending standards.

6.1. The role of bank-level adjustments

We assess the relative value of including bank-level information in the panel regressions used for the micro approach. Table 5 presents three regression specifications, each, respectively, with a broader set of controls. We provide a detailed description of the variables used as controls in the on line supplementary material.

The simplest specification (which we call Regression A) controls only for the lagged value of lending standards and for the change in loan demand reported in the SLOOS. The second specification (Regression B) includes those controls and additional macroeconomic variables that are likely to drive changes in lending standards. Finally, the third specification (Regression C) includes also a set of bank level controls. Our benchmark specification used Regression C (with the most controls) to construct our time series of adjusted lending standards.

A comparison of the values for the R-squared corresponding to each of the regressions shows that bank-specific variables do not add significant explanatory power. Indeed, the p-value for an exclusion test of the bank-specific variables is higher than 10 percent. In contrast, the exclusion test indicates that macroeconomic and aggregate financial variables are jointly significant in explaining the behavior of banks' lending standards.

We now use the residuals from each of the three regressions to construct three different time series of adjusted lending standards. Fig. 15 presents the impulse response functions under Alternative 1 using each of the three versions of adjusted lending standards. As

[17] Recall, also, that relative to Forni et al. (2024), our sample period is much shorter and at a quarterly frequency, while theirs is monthly.

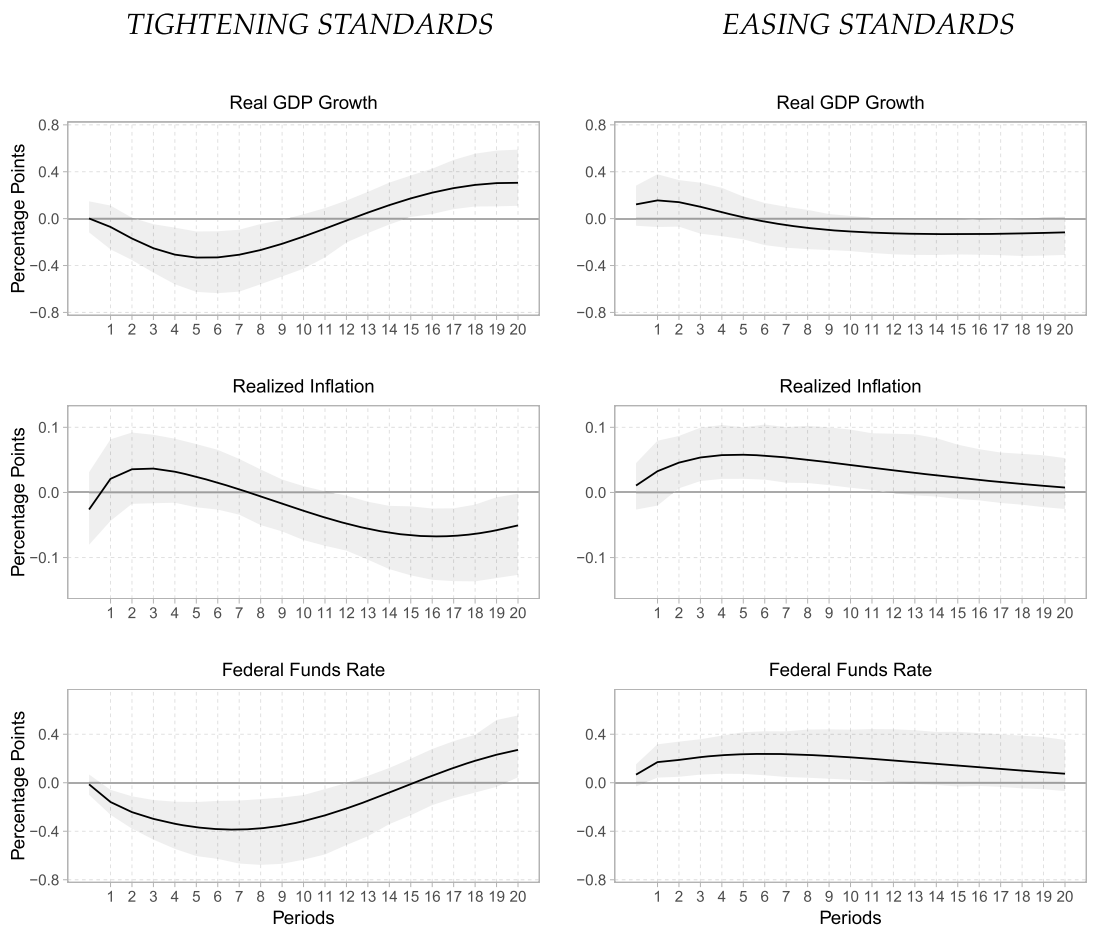


Fig. 13. Tightening and easing lending standards: OIRF based on overall lending standards
Notes: Using the VAR in Alternative 1, the plots correspond to the effect of a 1 SD positive (tightening) shock to lending standards (left column) and a 1 SD negative (easing) shock to lending standards (right column). The shaded area represents bootstrap 95% confidence intervals from 250 iterations. All models use 1 lags. The sample period is 1996:Q2 to 2019:Q4.

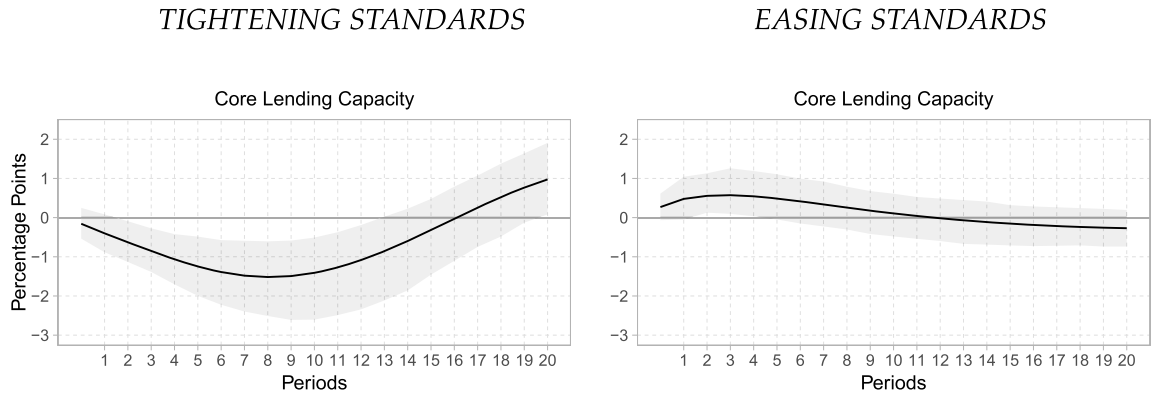


Fig. 14. Impact on bank core lending capacity of a shock to lending standards
Notes: Using the VAR in Alternative 1, the plots correspond to the effect on bank core lending capacity of a 1 SD positive (tightening) shock to lending standards (left column) and a 1 SD negative (easing) shock to lending standards (right column). The shaded area represents bootstrap 95% confidence intervals from 250 iterations. All models use 2 lags. The sample period is 1996:Q2 to 2019:Q4.

Table 5
Factors affecting changes in banks' overall lending standards (all banks).

Dependent variable: quarterly bank-level overall lending standards, ΔS_{it}						
	A		B		C	
	Est.	(S.E.)	Est.	(S.E.)	Est.	(S.E.)
Lagged Qtly. Change in Standards	0.563***	(0.016)	0.433***	(0.017)	0.387***	(0.019)
Qtly. Change in Loan Demand	-0.110***	(0.011)	-0.081***	(0.011)	-0.082***	(0.011)
Economic outlook: Real GDP			1.285	(0.830)	1.821*	(0.928)
Economic Outlook: Unemployment			17.123***	(1.751)	19.000***	(1.941)
Economic Outlook: 3-Month Treasury			-1.754	(1.234)	-1.647	(1.225)
Economic Outlook: 10-Year Treasury			-0.425	(2.264)	-1.094	(2.319)
Yearly Change in Real GDP			-1.498***	(0.372)	-1.397***	(0.424)
Yearly Change in Unemployment			1.199*	(0.646)	1.463*	(0.768)
Qtly. Change in EBP			8.577***	(1.156)	8.852***	(1.185)
Quarterly Change in Real FF Rate			-5.763***	(0.910)	-6.596***	(0.921)
Qtly. Change in VIX			-0.184*	(0.106)	-0.207*	(0.106)
Qtly. Change in NIMs					-2.275***	(0.815)
Qtly. Change in LLPs					1.460	(1.141)
Bank Size					-0.149	(1.136)
ROA					2.927*	(1.649)
Share of Core Loans					0.194**	(0.094)
Share of Core Deposits					0.068	(0.126)
Observations	5092		5092		5092	
$Pr > W^a$			0		0	
$Pr > W^b$					0.0616	
R^2	0.390		0.441		0.430	

Standard errors in parentheses
In the table, the symbol * stands for $p < .10$, ** stands for $p < .05$, and *** stands for $p < .01$
Notes: Sample period: 1996:Q1-2019:Q4; No. of banks = 91. The dependent variable in each specification is ΔS_{it}, which is the diffusion index of change in overall lending standards at bank i in quarter t. The variables aimed at capturing the economic outlook are defined as the expected four-quarter ahead change (i.e., $\mathbb{E}_{t-1}[y_{t+4} - y_t]$). Bank size is the log of total bank assets, and the share of core loans and deposits is expressed as a share of total bank assets. For the definition of other variables, see the data appendix. OLS estimates are under the column headings "Est." We report robust standard errors clustered at the bank level under the column headings "(S.E.)." All specifications include bank fixed effects (not reported).
The super-index "a" indicates the p-value for the exclusion test of common macroeconomic and financial explanatory variables; the super-index "b" indicates the p-value for the exclusion test of bank-specific explanatory variables.

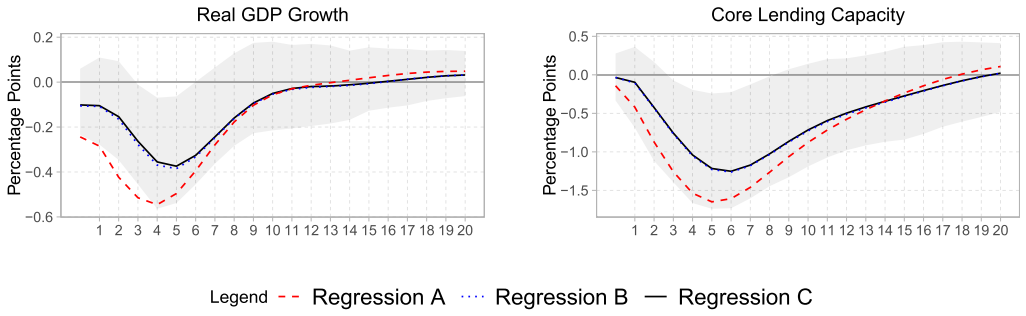


Fig. 15. Alternative bank-level adjustments: OIRF for a shock to overall adjusted lending standards
Notes: All three sets of impulse response functions are the result of implementing Alternative 1, each with a different definition of adjusted lending standards. The impulse functions are the reaction of each variable to a 1 SD shock to ΔS_t^a. The shaded area represents bootstrap 95% confidence intervals from 250 iterations, centered on the impulse response functions (in black) that result from using specification C of the panel regression. All VARs are estimated using 2 lags. The sample period is 1996:Q2 to 2019:Q4.

expected, the results based in regressions B and C produce very similar results, while the adjustment obtained with regression A gives results that suggest a stronger effect of standards on output and core lending capacity. Importantly, this suggests that the key source of endogeneity in lending standards is the influence of macroeconomic factors on banks' attitudes toward lending. Idiosyncratic factors, on the other hand, seem to play a much lesser role in aggregate outcomes.

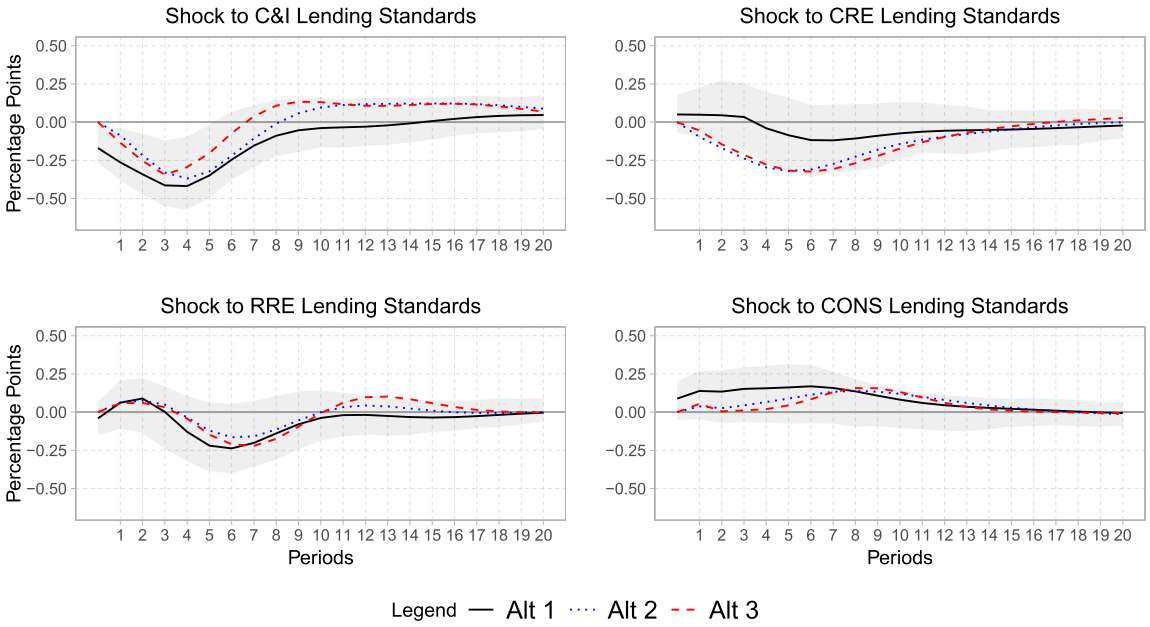


Fig. 16. Lending standards for the main loan categories: OIRF for real GDP growth in each case
Notes: The impulse response functions correspond to the effect on real GDP growth of a 1 SD shock to lending standards on C&I, CRE, RRE, and consumer loans. The shaded area represents bootstrap 95% confidence intervals from 250 iterations, centered on Alternative 1 (in black). All models use 2 lags. The sample period is 1996:Q2 to 2019:Q4.

6.2. Further disaggregation

Four Groups of Categories. We now consider four broad sets of categories of loans as in Cavallo et al. (2024a) and show how changes in lending standards for each of those four categories affect the macroeconomy. This disaggregation allows us to further disentangle the effects discussed in the previous sections. In particular, Fig. 16 shows that changes in standards for C&I loans are the main drivers behind the impact of business lending standards on GDP growth. While we only plot the impulse response function for output growth, results are similar in the case of inflation and interest rates.

Interestingly, when we use unadjusted lending standards in alternatives 2 and 3 to estimate our VAR, lending standards for CRE loans also appear to have meaningful effects on output. The lending standards on residential real estate and consumer loans do not impact output significantly regardless of the way we estimate their effects. These findings are in line with our conclusion that the main macroeconomic impact of changes in lending standards comes from their effects on business lending. The impulse response for core lending capacity of a lending-standards shock to each of the four main loan categories suggests again that the transmission of the shock goes from a drop in bank lending to lower GDP growth.

The Role of Large Banks. The U.S. banking industry is highly concentrated. With this in mind, we re-do our estimation procedures with a sub-sample of the SLOOS banks: the top 15 banks when ranked by asset size, in each quarter. Some of the banks rotate in and out of the sub-sample (and of the SLOOS sample, as well), making the data an unbalanced panel. Over our sample period, 22 different banks are at some point in the top 15 sub-sample, which reflects that rotation is not very prevalent.

The behavior of the index for adjusted lending standards using the full SLOOS sample is very similar to the index that only uses the top 15 largest banks in the sample. Indeed, this is the case even if we split the categories into the two natural supra-groups, lending to businesses and to households.

Perhaps not surprisingly, then, the effect on the macroeconomy is very similar when we consider the lending standards of the top 15 banks instead of the full SLOOS sample. A comparison of the impulse responses from a tightening of overall lending standards by the top 15 largest SLOOS banks and the one corresponding to the full sample of banks shows that most of the macroeconomic impact of a shock to lending standards comes from the activity of the largest banks. This is likely the result of focusing on aggregate effects: when assessing the impact of bank lending standards on GDP, inflation, and monetary policy, the activity of large banks is critical given their relative size.

Understanding the disparate role of smaller and larger banks is also important because shocks to the banking sector may not affect all banks equally. During the events that resulted in the liquidation of several regional U.S. banks in March 2023, deposits flowed from smaller to larger banks, giving larger banks a more favorable position to possibly increase lending, while smaller banks were experiencing the opposite.[18]

[18] Cavallo et al. (2024b) discuss in more detail the recent evolution of bank lending standards.

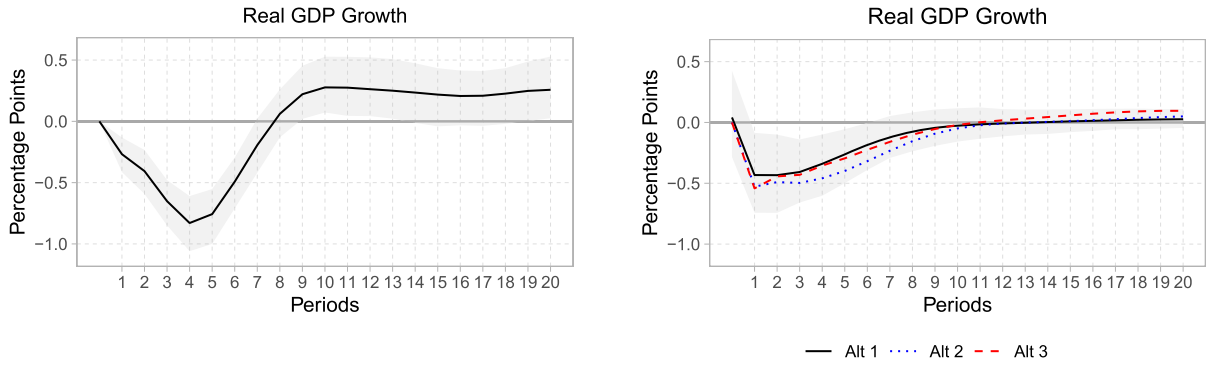


Fig. 17. Alternative sample periods: OIRF for a tightening shock to overall lending standards
Notes: The left panel goes from 1991:Q3 to 2019:Q4, excludes the years of the GFC (2007 to 2011), and uses bank lending standards only for C&I. The sample period for the right panel goes from 1996:Q2 to 2023:Q4 and uses our measure of overall lending standards. The shaded area represents bootstrap 95% confidence intervals from 250 iterations. All models are estimated using 2 lags.

With this in mind, we also run a VAR where we separate standards for the top 15 banks and for the rest of the banks, but we include them both in the estimation. While lending standards for the two groups often move together, the correlation during our sample period is only 0.66, so there is some differential variation. We find that a shock to the lending standards of those banks outside of the top 15 indeed has meaningful effects on output, independently of the shock to lending standards of the top 15.

6.3. *Sample-period robustness*

Excluding the Global Financial Crisis. Our sample period includes the GFC, an event that could have produced extreme versions of the dynamics we identify in our study. It is then natural to consider the possibility that the intensity of our results could be, effectively, capturing outlier observations occurring at that time. To assess this possibility, we re-estimate our benchmark specifications excluding the years of the GFC.

The GFC was a protracted episode and removing it from the data leaves us with a smaller sample. For this reason, we proceed to extend the sample period back to the second half of 1991; i.e., roughly five more years. Computing overall lending standards (the way we define them) is only possible for the period starting in 1996. However, lending standards for C&I can be computed from the third quarter of 1991 onward. Hence, we proceed to estimate our Alternative 2, in expression 5, using C&I lending standards for a sample that goes from 1991 to 2019, but excludes the GFC years, 2007 to 2011.[19]

The left panel of Fig. 17 presents the impulse response function for GDP growth of a tightening shock to C&I lending standards. The model is tightly identified, and its behavior is very consistent with our benchmark specification. If anything, the impact of lending standards on GDP growth is stronger when excluding the GFC. More generally, we have confidence that the effects we uncover represent a general pattern in our data.

Including the COVID Period. So far, we have left the period of the COVID pandemic out of the estimation, as this was a time when all macroeconomic variables were strongly impacted by the economic crisis that ensued. Our results, however, do not change dramatically when taking that period into account.

The right panel of Fig. 17 presents the impulse response function for GDP growth of a tightening shock to overall lending standards, estimated using the sample period that extends until the end of 2023 and hence includes the years of the pandemic. While the impact on GDP growth of a shock to overall lending standards appears larger when including the COVID times, the accuracy of the estimation represented by the confidence intervals is lower. This makes sense given the significant noise introduced in the estimation from including the pandemic years; multiple unrelated effects affected macroeconomic outcomes during that time. We conclude that our findings are generally robust to including the COVID episode.

6.4. *Estimation robustness*

Local Projections. To compare with much of the previous literature and to be able to assess the implications of pursuing a micro or macro approach in controlling for endogeneity, we have used standard VARs for most of our analysis so far, complemented with timing restrictions for identification. While our estimates are minimally sensitive to the specific number of lags used in the VARs, it is still possible that when we use the adjusted lending standards in the estimation, the impulse responses we obtain from a VAR could be biased (Li and Wolf, 2024). With this in mind, we follow Cavallo et al. (2024a) and use *local projections* to assess the robustness of our findings.

[19] We use Alternative 2 because it is the most parsimonious, less data-demanding alternative. Alternative 3 gives comparable results.

Using local projections (Jordà, 2005) for estimation we find that the impact of a tightening shock to adjusted lending standards on output growth is in line with the results in Fig. 3: after a one standard deviation tightening shock, output growth declines as much as 0.6 percent, with the peak effect five quarters after the shock. Furthermore, the estimates using local projection display a very clear disparity between the effect of a shock to business lending standards and a shock to households lending standards, with most of the effect on output coming from changes in lending standards to businesses. This, again, supports the robustness of our results using VARs.

Identification and the Ordering of Variables in the VAR. Following Gilchrist and Zakrajšek (2012), the study by Mumtaz et al. (2018) advocates for a different ordering of variables in the VARs. In particular, the alternative order allows interest rates to react contemporaneously to credit shocks. To assess the impact on our findings of variations in the identification assumptions, we consider a VAR that uses the same variables as in Alternative 2 but under the following order: output growth, core lending capacity growth, inflation, *changes in overall lending standards*, bond credit spreads, and the real fed funds interest rate.

The alternative ordering permits contemporaneous effects of the shock to standards on the fed funds rate and the corporate bond spread. These contemporaneous effects are statistically significant, but they are small in an economic sense. In general, all impulse response functions under the new ordering of variables are very similar to those under the benchmark ordering.

7. Conclusion

In this paper, we aimed to better understand the role of bank lending standards in the U.S. economy. We produced three main findings: one methodological and the other two about the economic mechanisms at play in the data. Methodologically, we find that using micro data to adjust for endogeneity produces relatively small differences when comparing with the more traditional macro VAR approach. This finding is important because, in many instances, the bank-level data is not readily available to the researcher. Our study suggests that this should not be regarded as an insurmountable limitation for those interested in the macroeconomic side of the issues.

One clear advantage of the adjustment of bank lending standards using micro data is the possibility of having extra information (bank-level cross-sectional data) to address the questions at hand. This is especially valuable given the limited data that can be brought to bear in this context. In particular, good, comprehensive, and consistent data on aggregate lending standards are relatively recent. For the purpose of this paper, we can go back only to 1996, for example, given our interest in high-level disaggregation by categories of credit. However, since then, the U.S. economy has experienced only three main business cycles, and the interaction between the financial system and the economy is heavily influenced by the events in 2008, which was a very special historical episode.

The lack of an extensive sample period also limits the number of variables that can be included in a VAR without compromising accuracy in the estimation. In principle, as the time series on standards becomes longer and the sample period covers a broader set of macroeconomic outcomes, the VAR approach to the identification of lending-standards shock can be counted to produce reliable results. In the meantime, however, using the micro data to complement the analysis, to the extent possible (given that some of the data remain confidential), is a valuable avenue to consider.

In terms of the economics captured by our data, we see that a *tightening* of lending standards can have significant negative impact on output growth, with the peak effect about five quarters after the shock. We also see that this impact is much larger when standards tighten for business lending, relative to when they tighten for household lending. Furthermore, much of the effect captured in the aggregate data comes from the impact that a tightening of lending standards has on bank lending capacity and real GDP when interest rates are away from their effective lower bound, when output is below its long-run trend, and when financial uncertainty is increasing or high. If, instead, lending standards are easing, or output is above trend, or interest rates are close to their lower bound, or uncertainty is decreasing or low, then lending standards do not appear to have significant effects on economic activity.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Supplementary material

Supplementary material associated with this article can be found, in the online version, at 10.1016/j.jedc.2025.105247.

References

Abadi, J., Brunnermeier, M., Koby, Y., 2023. The reversal interest rate. Am. Econ. Rev. 113 (8), 2084 2120.
Ajello, A., 2016. Financial intermediation, investment dynamics, and business cycle fluctuations. Am. Econ. Rev. 106 (8), 2256–2303.
Alfaro, L., García-Santana, M., Moral-Benito, E., 2021. On the direct and indirect real effects of credit supply shocks. J. Financ. Econ. 139 (3), 895–921.
Barnichon, R., Matthes, C., Ziegenbein, A., 2022. Are the effects of financial market disruptions big or small? Rev. Econ. Statist. 104 (3), 557–570.
Baslandze, S., Minoiu, C., Penciakova, V., Willis, J.L., 2023. Do credit supply shocks constrain employment growth of small and medium-sized enterprises? Atlanta Fed. Policy Hub. 5.
Bassett, W.F., Chosak, M.B., Driscoll, J.C., Zakrajšek, E., 2014. Changes in bank lending standards and the macroeconomy. J. Monet. Econ. 62, 23–40.
Baumeister, C., Hamilton, J.D., 2015. Sign restrictions, structural vector autoregressions, and useful prior information. Econometrica 83 (5), 1963–1999.
Bemanke, B., Gertler, M., 1989. Agency costs, net worth, and business fluctuations. Am. Econ. Rev. 79 (1), 14–31.
Bernanke, B.S., 1983. Nonmonetary effects of the financial crisis in the propagation of the great depression. Am. Econ. Rev. 73 (3), 257–276.

Brunnermeier, M.K., Eisenbach, T., Sannikov, Y., 2013. Macroeconomics with financial frictions: a survey. Advances in Economics and Econometrics: Tenth World Congress *Vol. II: Applied Economics* , 3–94.
Brunnermeier, M.K., Palia, D., Sastry, K.A., Sims, C.A., 2021. Feedbacks: financial markets and economic activity. Am. Econ. Rev. 111 (6), 1845-1879.
Buera, F.J., Nicolini, J.P., 2020. Liquidity traps and monetary policy: managing a credit crunch. Am. Econ. J.: Macroecon. 12 (3), 110–138.
Castro, A., Glancy, D., Ionescu, F., 2022a. Drivers of bank supply of business loans. FEDS Notes, Board of Governors of the Federal Reserve System (February 22).
Castro, A., Glancy, D., Ionescu, F., Marchal, G., 2022b. What happens when banks tighten c&i loan supply? FEDS Notes, Board of Governors of the Federal Reserve System (February 18).
Cavallo, M., Morelli, J., Zarutskie, R., 2024a. Unpacking the effects of bank credit supply shocks on economic activity. FEDS Notes, Board of Governors of the Federal Reserve System (May 24).
Cavallo, M., Morelli, J., Zarutskie, R., Baylor, S., 2024b. Measuring bank credit supply shocks using the senior loan officer survey. FEDS Notes, Board of Governors of the Federal Reserve System (May 24).
Chodorow-Reich, G., 2014. The employment effects of credit market disruptions: firm-level evidence from the 2008-09 financial crisis. Q. J. Econ. 129 (1), 1–59.
Chodorow-Reich, G., Darmouni, O., Luck, S., Plosser, M., 2022. Bank liquidity provision across the firm size distribution. J. Financ. Econ. 144 (3), 908–932.
Eggertsson, G.B., Juelsrud, R.E., Summers, L.H., Wold, E.G., 2024. Negative nominal interest rates and the bank lending channel. Rev. Econ. Stud. 91 (4), 2201–2275.
Forni, M., Gambetti, L., Maffei-Faccioli, N., Sala, L., 2024. Nonlinear transmission of financial shocks: some new evidence. J. Money Credit Bank. 56 (1), 5–33.
Gambetti, L., Musso, A., 2017. Loan supply shocks and the business cycle. J. Appl. Econometr. 32 (4), 764–782.
Gilchrist, S., Zakrajšek, E., 2012. Credit spreads and business cycle fluctuations. Am. Econ. Rev. 102 (4), 1692–1720.
Glancy, D., Kurtzman, R., Zarutskie, R., 2020. An aggregate view of bank lending standards and demand. FEDS Notes, Board of Governors of the Federal Reserve System (May 4).
Hamilton, J.D., 2018. Why you should never use the Hodrick-Prescott filter. Rev. Econ. Statist. 100 (5), 831–843.
Ivashina, V., Laeven, L., Moral-Benito, E., 2022. Loan types and the bank lending channel. J. Monet. Econ. 126, 171–187.
Jimenez, G., Ongena, S., Peydro, J.L., Saurina, J., 2012. Credit supply and monetary policy: identifying the bank balance-sheet channel with loan applications. Am. Econ. Rev. 102 (5), 2301–2326.
Jiménez, G., Mian, A., Peydró, J.-L., Saurina, J., 2020. The real effects of the bank lending channel. J. Monet. Econ. 115, 162–179.
Jordà, O., 2005. Estimation and inference of impulse responses by local projections. Am. Econ. Rev. 95 (1), 161–182.
Kilian, L., Lütkepohl, H., 2017. Structural Vector Autoregressive Analysis. Cambridge University Press.
Kilian, L., Murphy, D.P., 2012. Why agnostic sign restrictions are not enough: understanding the dynamics of oil markets VAR models. J Eur Econ Assoc 10 (5), 1166–1188.
Lenza, M., Primiceri, G.E., 2020. How to estimate a VAR after March 2020. National Bureau of Economic Research Working Paper 27771 .
Li, Dake, P.-M.M., Wolf, C.K., 2024. Local projections vs. VARs: lessons from thousands of DGPs. J. Econom. (forthcoming).
Lown, C., Morgan, D., 2006. The credit cycle and the business cycle: new findings using the loan officer opinion survey. J. Money, Credit Bank. 38 (6), 1575–1597.
Ludvigson, S.C., Ma, S., Serena, N., 2021. Uncertainty and business cycles: exogenous impulse or endogenous response? Am. Econ. J.: Macroecon. 13 (4), 369–410.
McCord, Roisin, Prescott, Edward, S., 2014. The financial crisis, the collapse of bank entry, and changes in the size distribution of banks. *Fed. Reser. Bank Richmond* Econ. Q. 100 (1), 23–50.
Mumtaz, H., Pinter, G., Theodoridis, K., 2018. What do VARs tell us about the impact of a credit supply shock? Int. Econ. Rev. (Philadelph.) 59 (2), 625–646.
Plagborg-Møller, Mikkel, Wolf, Christian, K., 2021. Local projections and VARs estimate the same impulse responses. Econometrica 89, 955–980.
Ramey, V.A., 2016. Macroeconomic shocks and their propagation. Handb. Macroecon. 2, 71–162.
Stiglitz, Joseph, Weiss, Andrew, 1981. Credit rationing in markets with imperfect information. Am. Econ. Rev. 71 (3), 393–410.
Swarbrick, J., 2023. Lending standards, productivity, and credit crunches. Macroecon. Dyn. 37, 456–481.
Uhlig, H., 2004. Do thechnolgy shocks lead to a fall in total hours worked? J. Eur. Econ. Assoc. 2 (2-3), 361–371.
Williamson, S., 1987. Costly monitoring, loan contracts, and equilibrium credit rationing. Q. J. Econ. 102 (1), 135–145.